

Airborne Electromagnetic Characterization of Shallow Permafrost in the Canadian Arctic for Infrastructure Challenges

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From: Quito, Ecuador

- **Colorado School of Mines, Golden CO, USA (May 2024):** B.Sc. in Geophysical Engineering, Minor: Data Science
- **University of British Columbia (UBC), Vancouver, BC, Canada:** M.Sc. in Geophysics (Thesis-based), Geophysical Inversion Facility

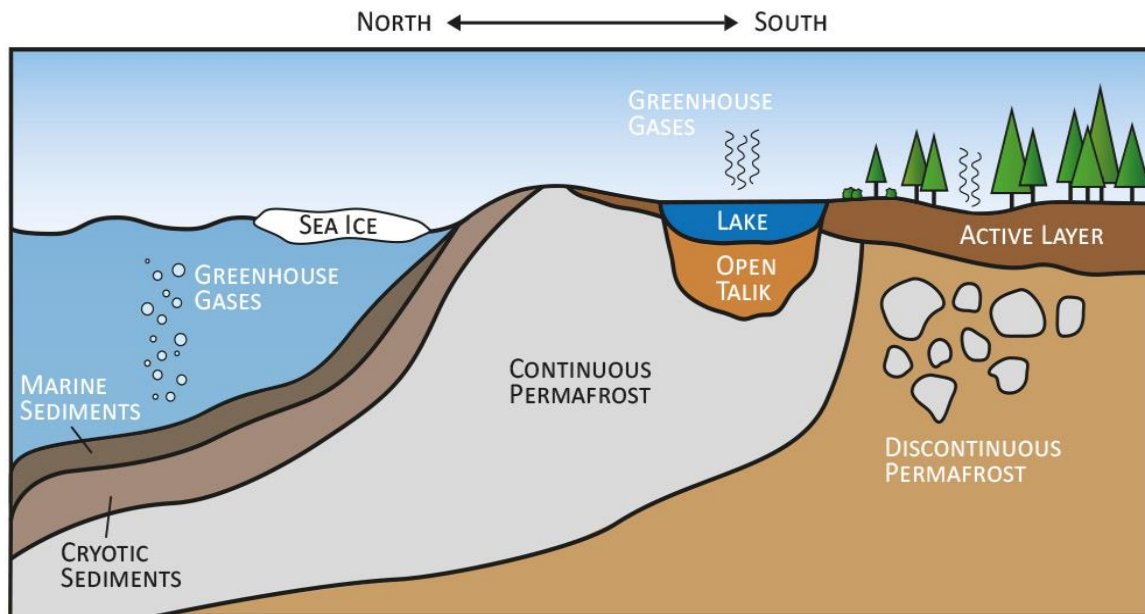
Some Relevant Experiences:

- **Teck Resources Geophysics Co-op** (Summer 2025): Supported the Exploration Geophysics team.
- **Geoscience Gitsegukla** (Sep 2025–Present): Developing geoscience education and geophysical exploration materials for Indigenous-led learning initiatives in Northern BC.

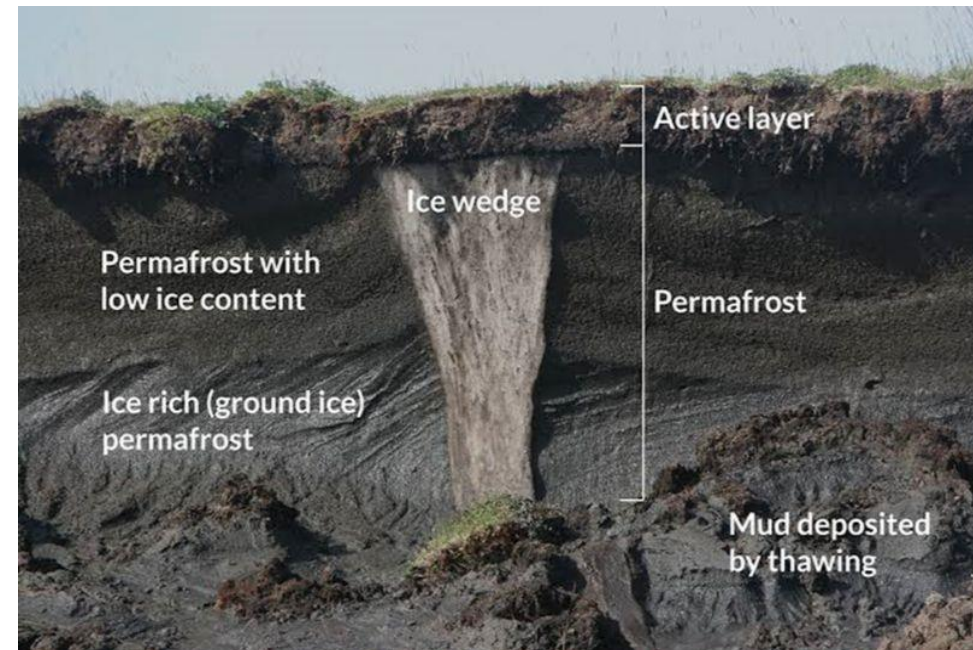
Research Interests: Electromagnetic methods, Machine Learning, Integration of remote sensing data into geophysics, Energy, Infrastructure applications

Permafrost?

- ▶ **Permafrost:** Any ground (soil, rock, or sediment) that stays completely frozen at or below 0°C for at least two years in a row
- ▶ **Active Layer:** Top layer of ground above the permafrost that thaws in the summer and refreezes in the winter.
- ▶ **Ice Rich Layer:** Specific part of the permafrost that contains a very high volume of actual ice.



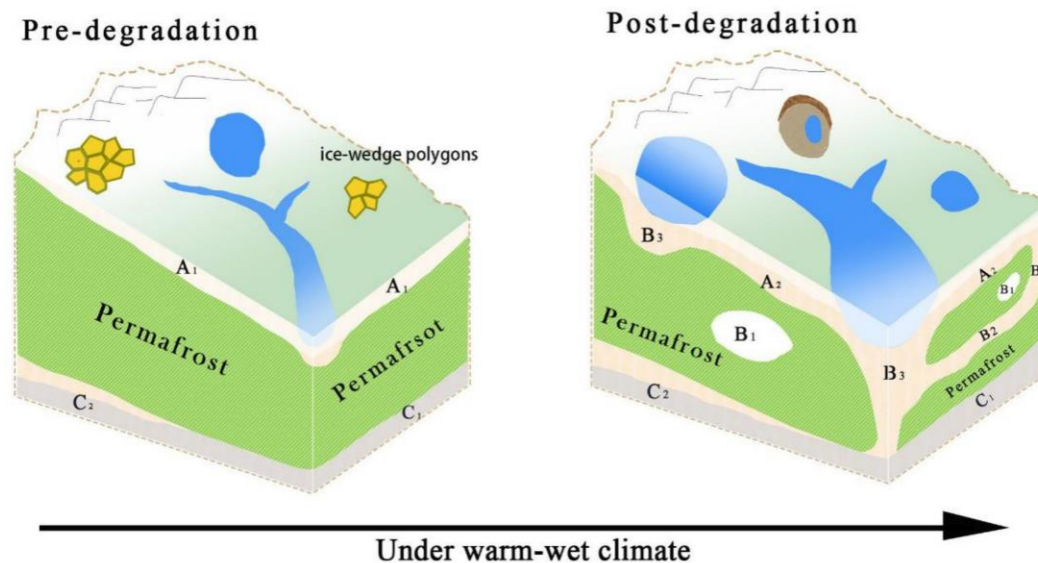
(Global Tipping Points Report, 2023)



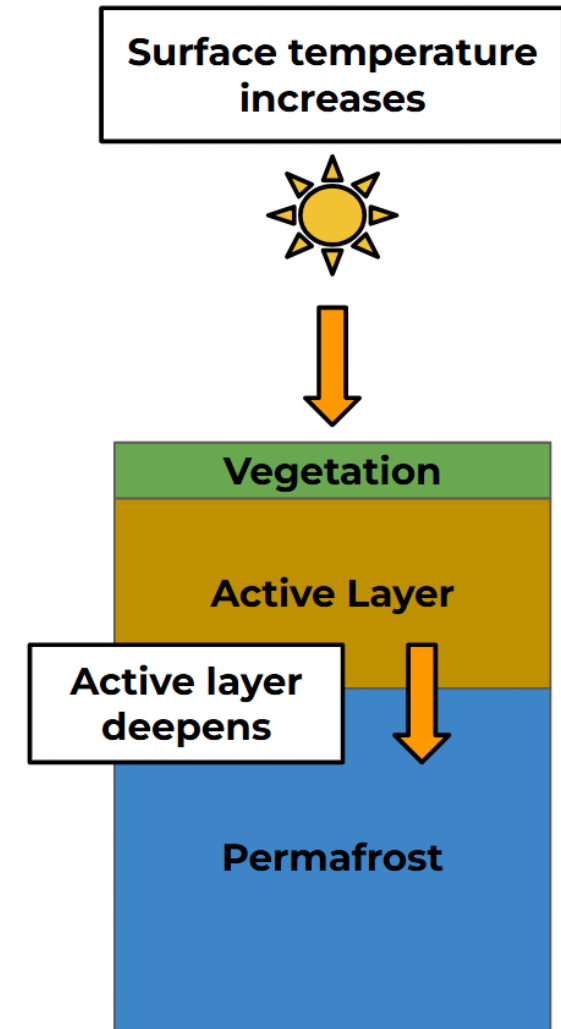
(Badamasi, 2022)

Permafrost degradation

- ▶ **Climate change accelerating permafrost degradation**, thickening the active layer, and increasing water content (surface and subsurface).
- ▶ **Ice-rich permafrost degrades more severely** — producing more meltwater and greater ground instability upon thaw.



(Jin et al., 2022)



Permafrost degradation and infrastructure

Increased water flow + ground softening → threaten infrastructure stability

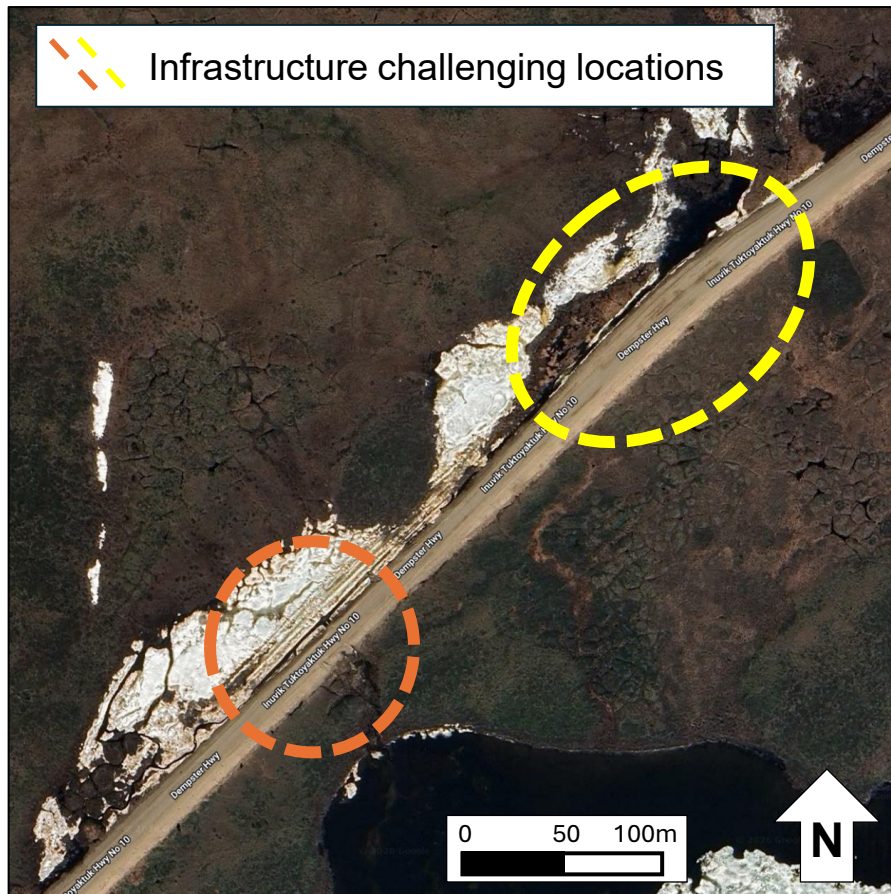
- Requires sustained subsurface monitoring and characterization



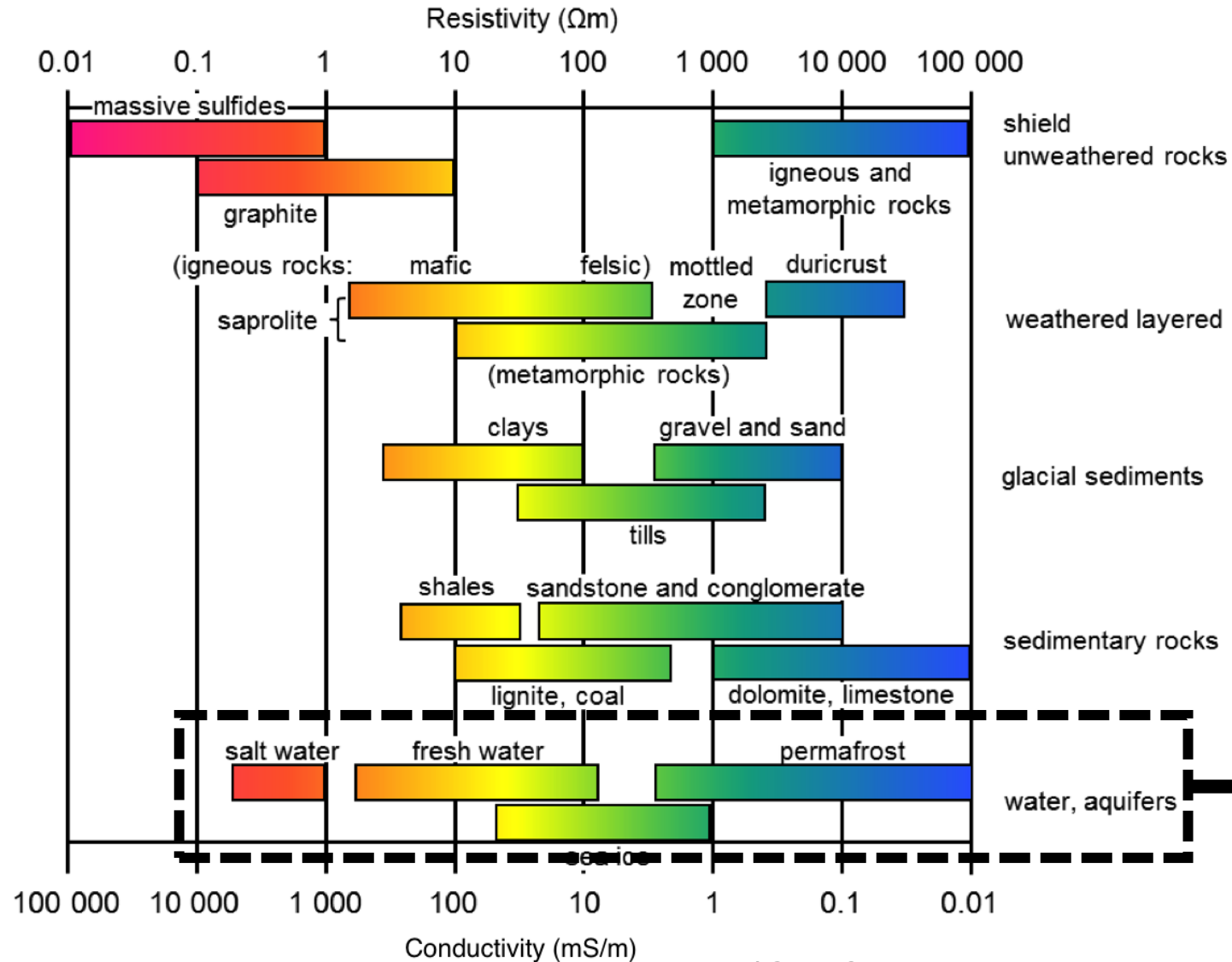
Infrastructure problem focus

Identify areas of faster permafrost degradation — via signatures of thaw, water flow, or ice-rich zones — since these pose the greatest infrastructure risk.

Pictures from locations



Resistivity and permafrost

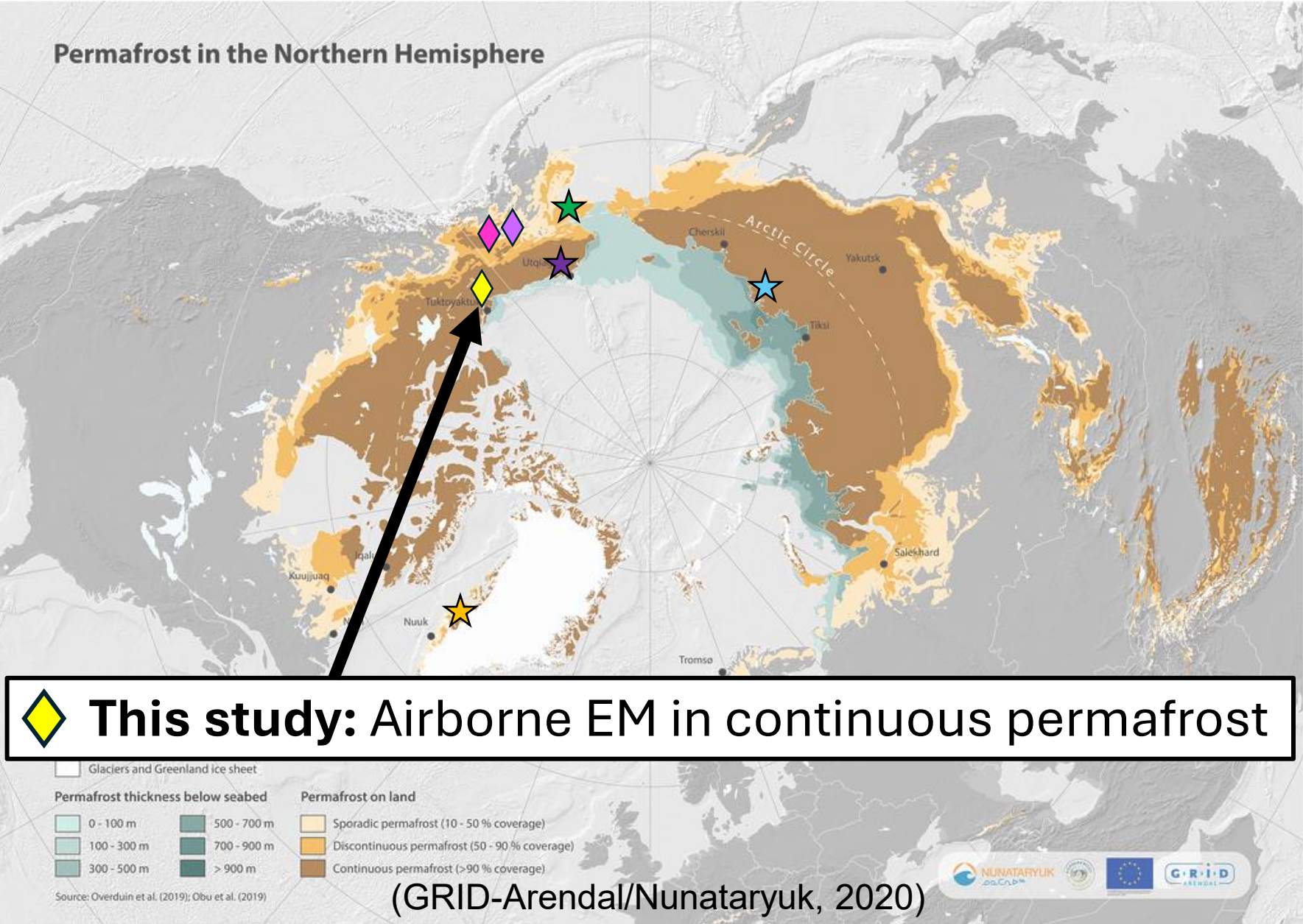


Changes in resistivity are primarily controlled by water phase (frozen vs. unfrozen).



(GeoSci.xyz Developers, 2017)

Prior EM-based permafrost investigations



◊ Airborne EM

☆ Ground EM

☆ Jones et al., 2023, GRL; Rangel et al., 2021, JGR; Creighton et al., 2018, JGR

☆ Sullivan et al., 2021, GRL

☆ Yakovlev et al., 2018, Earths Cryosphere

☆ Lorentzen et al., 2024, Geophysics

◊ Walter Anthony et al., 2018, Nature Comms

◊ Minsley et al., 2012, GRL; Rey et al., 2019, ERL

*Content adapted from Gustafson's 2026 EMIW presentation

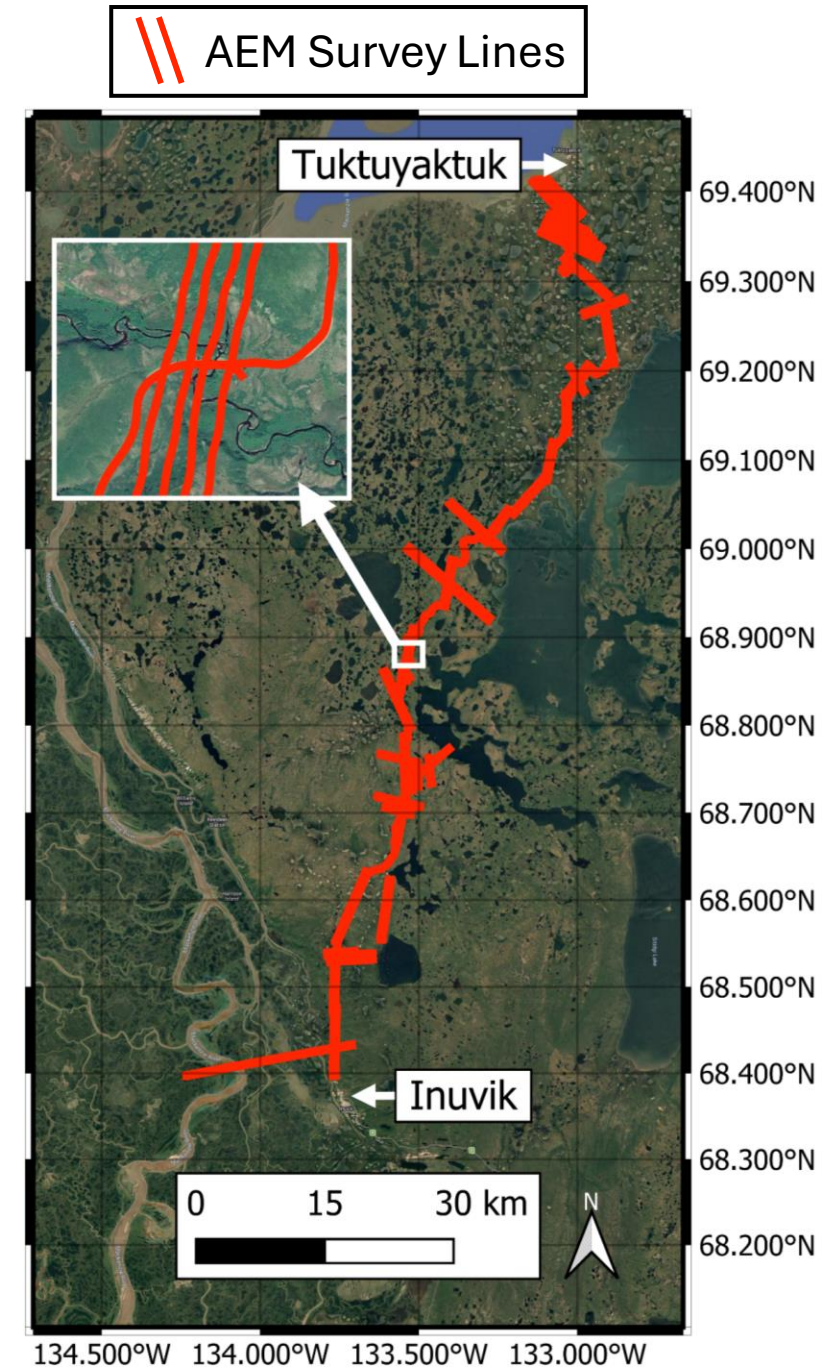
Setting – Canadian Arctic highway

- ▶ Beaufort delta region of the **Northwest Territories**, most ice rich permafrost landscape in North America (Kokelj et al., 2023).
- ▶ **Inuvik – Tuktoyaktuk Highway (ITH)**, 138 km all-season highway built on permafrost.
- ▶ Critical role for **healthcare, education, and economic opportunities** for the northern communities.



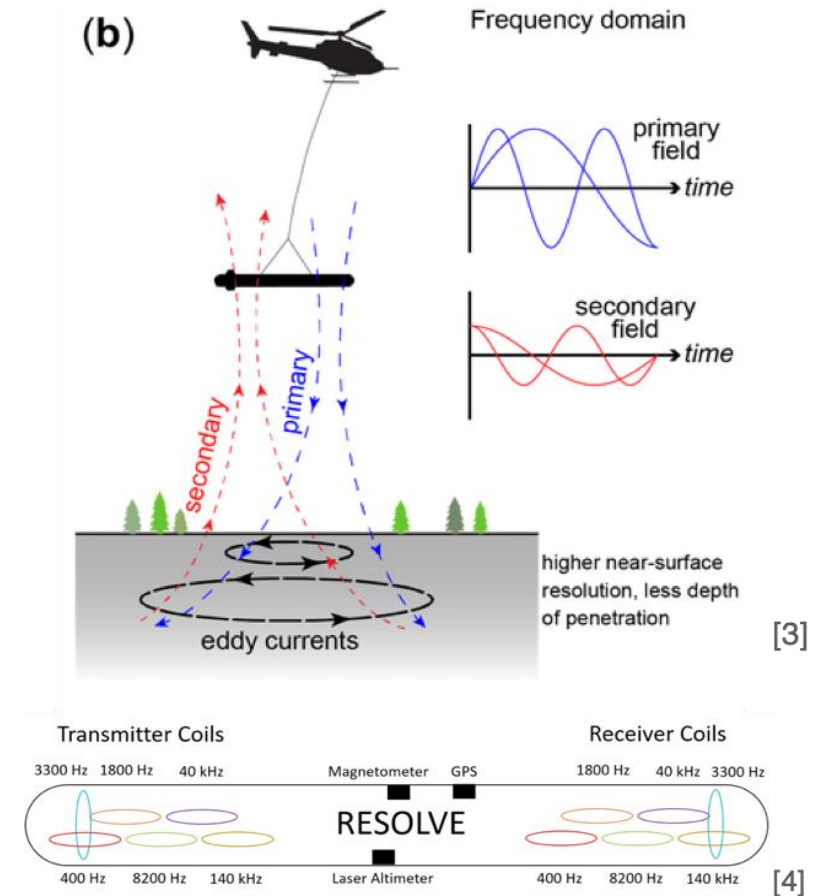
Objective and Method

- ▶ Use Airborne Electromagnetics (AEM) for mapping permafrost geometry over large regions.
- ▶ Focus on the shallow subsurface: active-layer, ice content, and seasonal thaw dynamics — to map faster-degrading zones and support infrastructure planning.

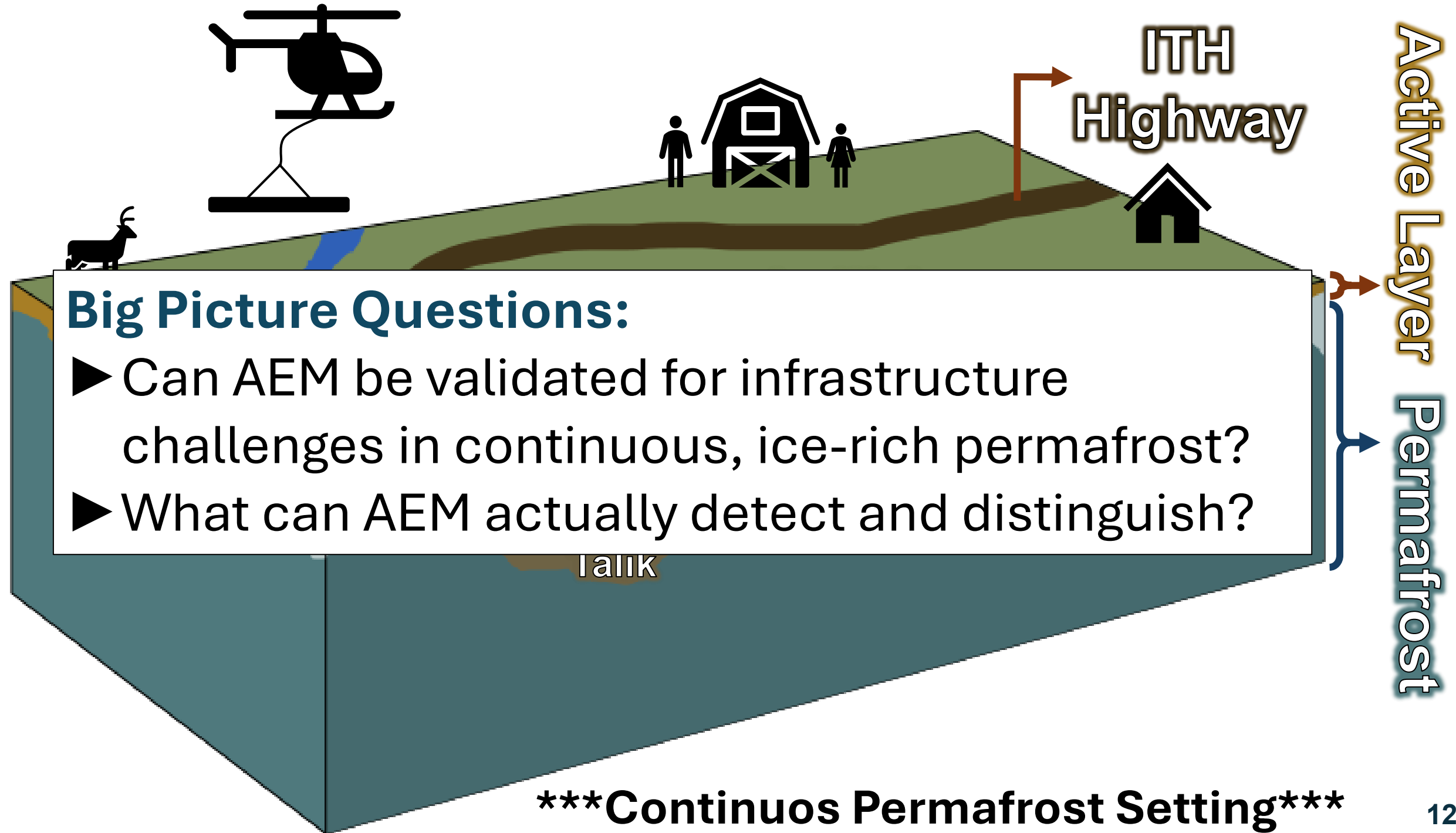


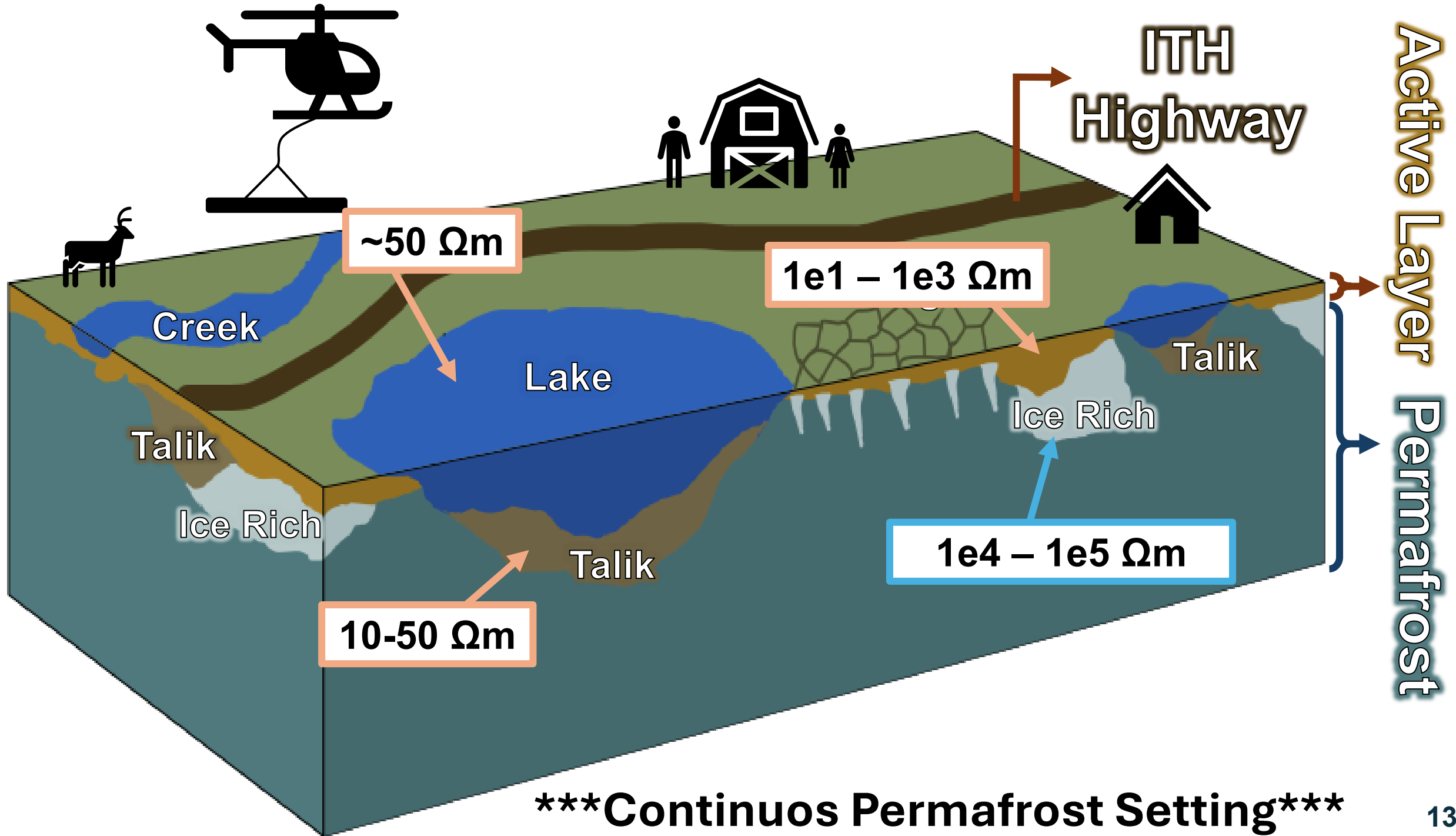
Airborne Frequency-Domain Electromagnetic (AEM)

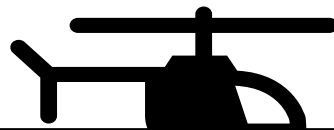
Survey System	RESOLVE
Operator	Xcalibur Geophysics
Date	August 2024
Frequencies	Coplanar: 135 kHz, 40 kHz, 8.2 kHz, 1.8 kHz, and 400 Hz, Coaxial 3.3 Hz.
Flight Height	35 m
Line Spacing	100 m
Total Line Km	1,142 km



- ▶ AEM as a spatially extensive, time-efficient, cost-effective compared to traditional regional permafrost characterization.
- ▶ **Longest survey of its kind for permafrost applications**

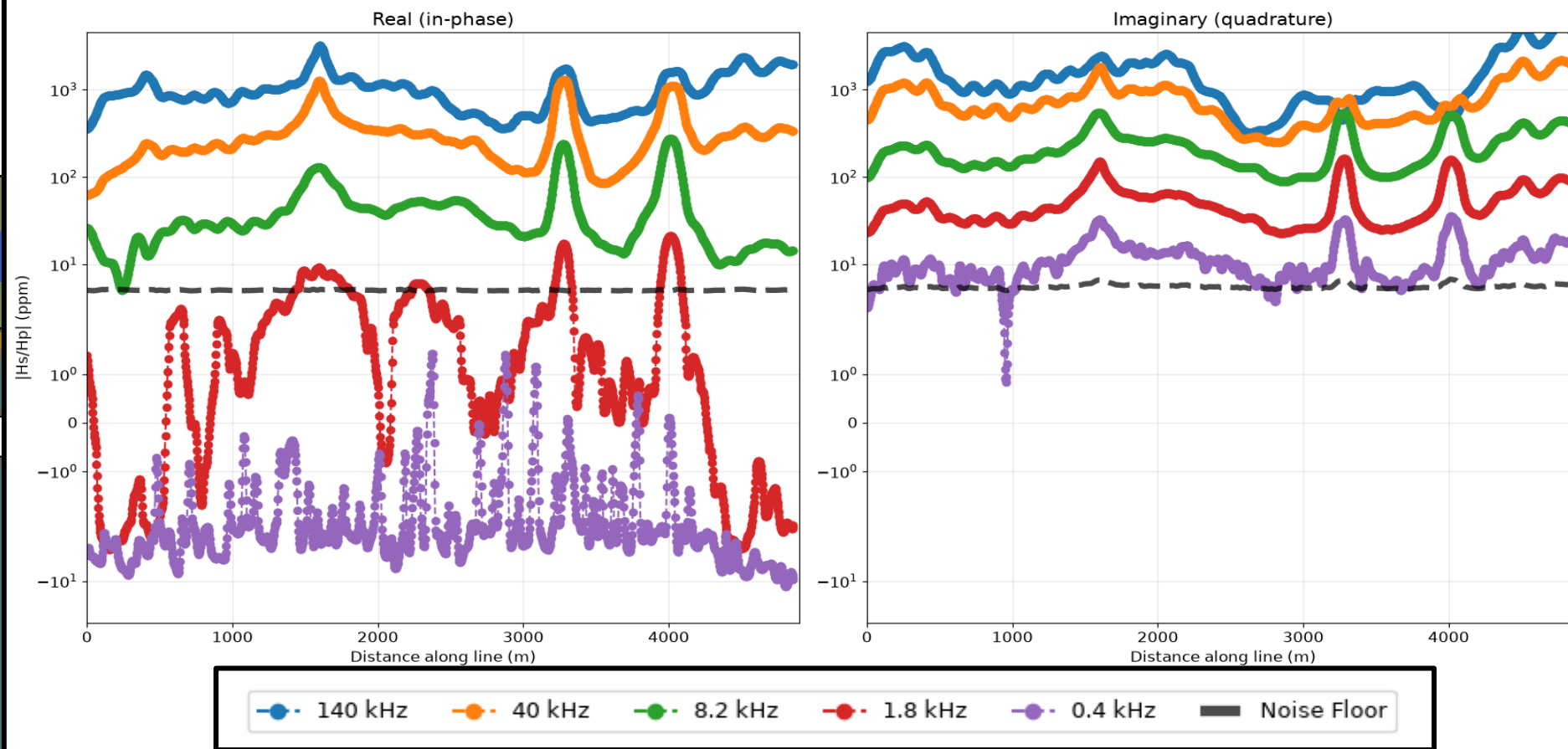






ITH

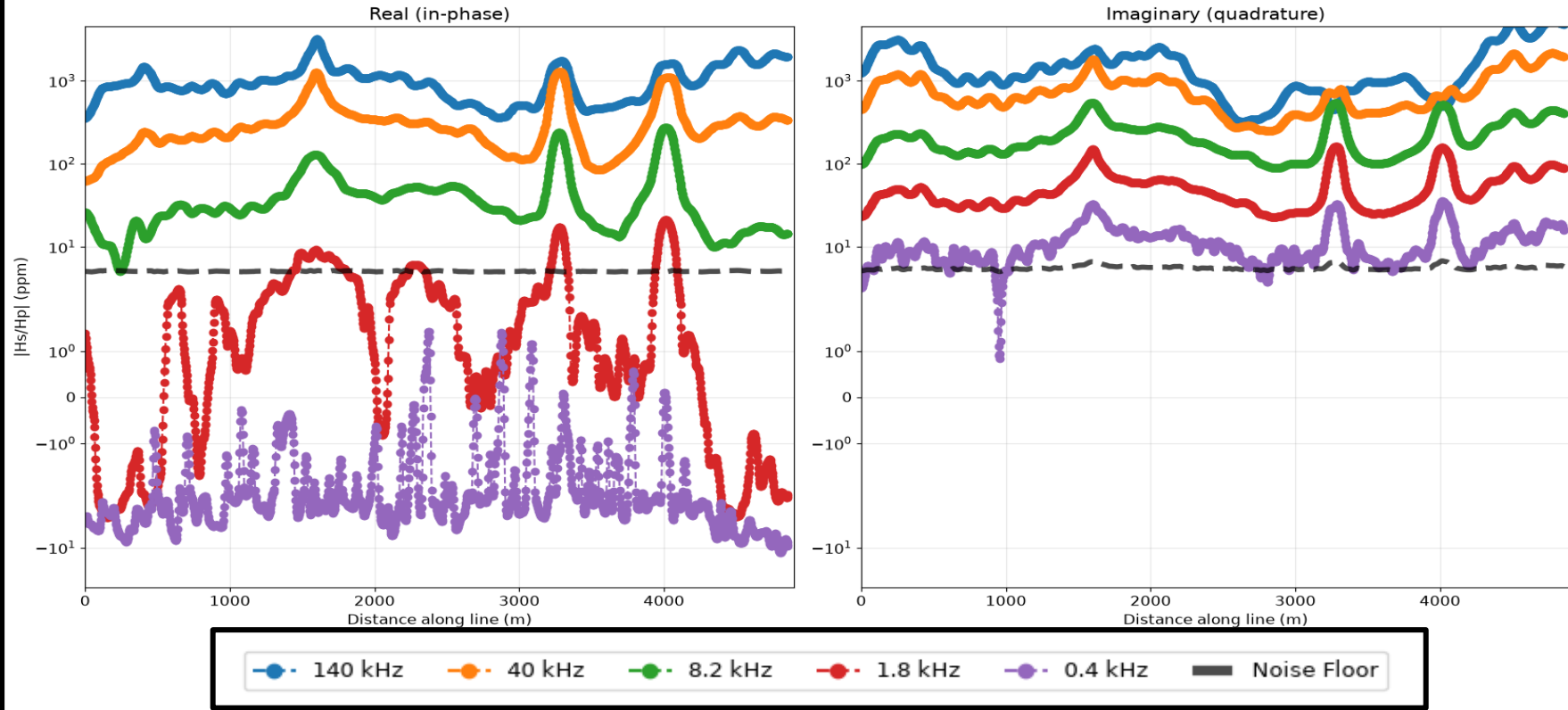
Example of data gathered



Active Layer Permafrost

Continuous Permafrost Setting

Example of data gathered



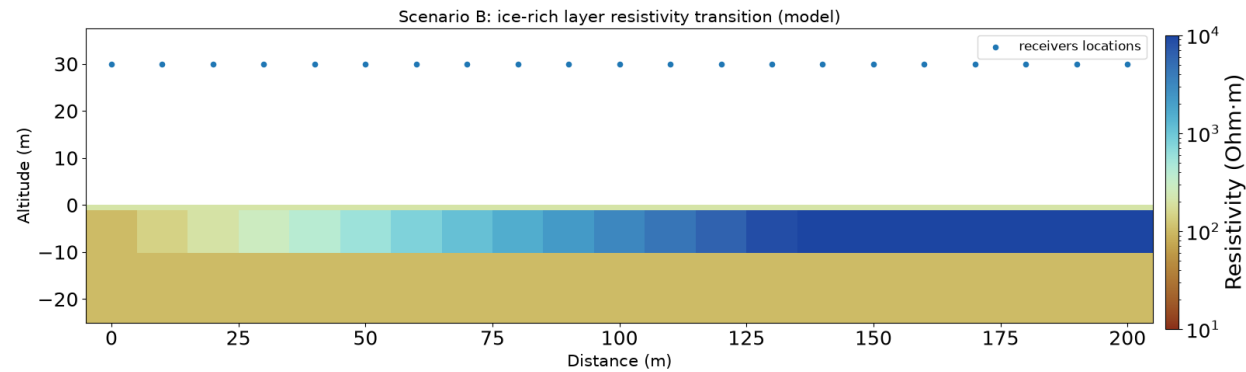
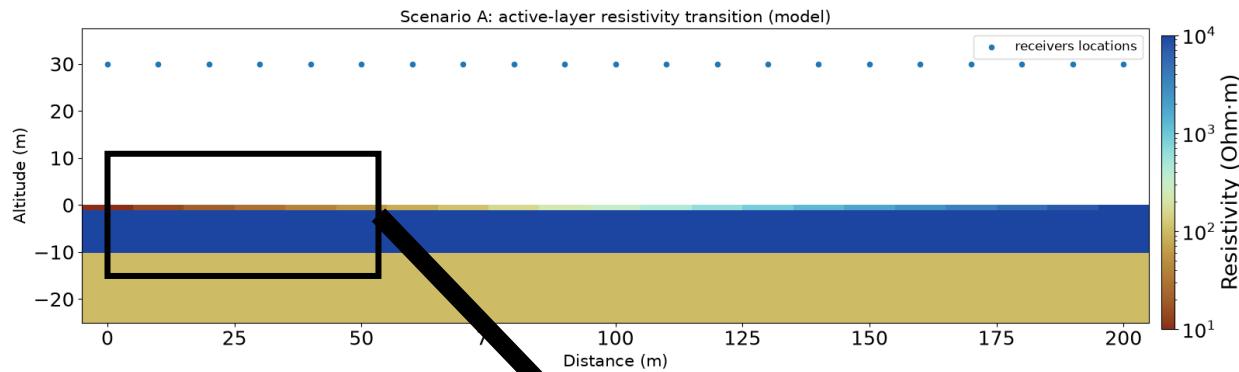
Main data challenges:

- Poor low-frequency data at higher latitudes (below noise floor), where permafrost is more resistive
- Abrupt lateral changes, highly conductive to highly resistive along flight lines

AEM Simulations, Effect of Changing “Active Layer” vs Changing “Ice Rich Layer”

Changing Active Layer (1 m):
10 to 1000 Ohm*m

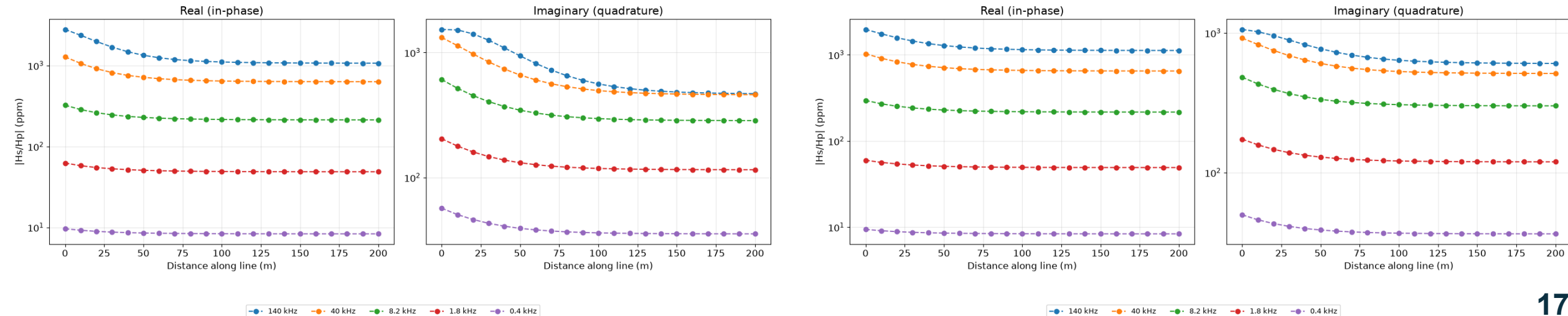
Changing Ice Rich Layer (10 m):
100 to 100'000 Ohm*m



Effect of Changing “Active Layer” vs Changing “Ice Rich Layer”?

AEM Simulations observations:

- ▶ **Shallow-layer changes influence AEM response as strongly as deeper ice-content variations**
- ▶ **Low-frequency data shows little variation — consistent with deeper resistive structures staying relatively continuous in this resistive setting**
- ▶ **Partly explains why some of the lowest frequencies having limited usable data (values fall below the noise floor)**



Inversion Workflow

Quasi-2D models built by stitching together 1D inversions

Workflow requirements:

- ▶ **Focus mainly on shallow structure:** prioritize high-frequency data (more consistent, usable data)
- ▶ **Need a sounding-specific a priori model**, not a single half-space for a whole line — biases each sounding's results and loses shallow value/thickness
- ▶ **Most structure changes sits in the shallow zone (~30 m);** deeper resistive zones are expected to be smoother

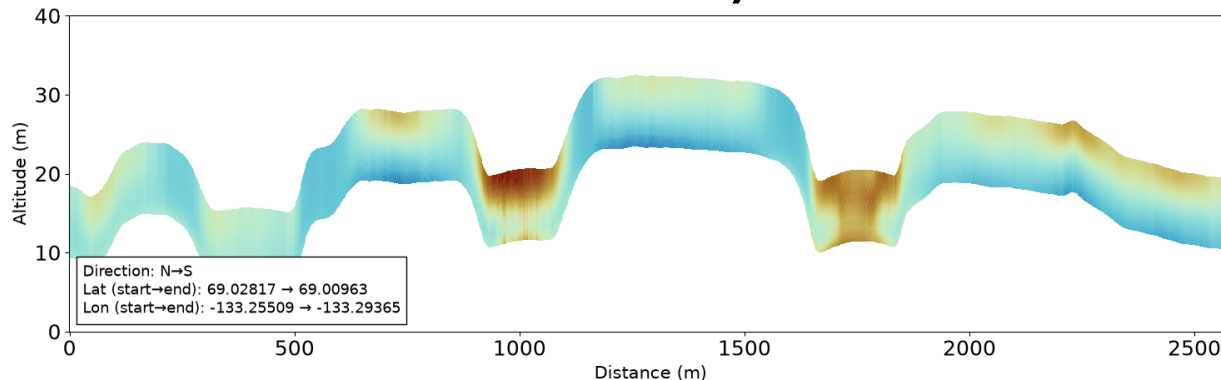
Inversion Workflow

1. Building the shallow-layer-informed starting/reference model

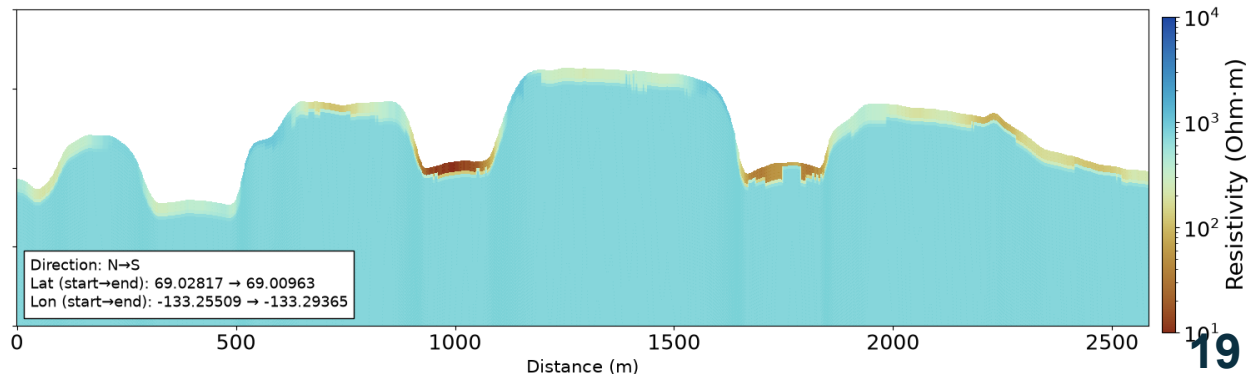
- a) **Shallow L2 inversion**, for each sounding using highest frequencies only (140 kHz, 40 kHz)
- b) **Build the shallow (active) layer**: for each sounding, detect the layer depth at the point of highest resistivity gradient, then set the layer's value to the average resistivity above that depth.
- c) **Background/basement resistivity**: from a quick 1-layer half-space inversion on select soundings (lowest usable frequency)

→ **Combine**: b) as top layer + c) as bottom layer → **starting/reference model**

Part a)



Starting/reference model: b) + c)



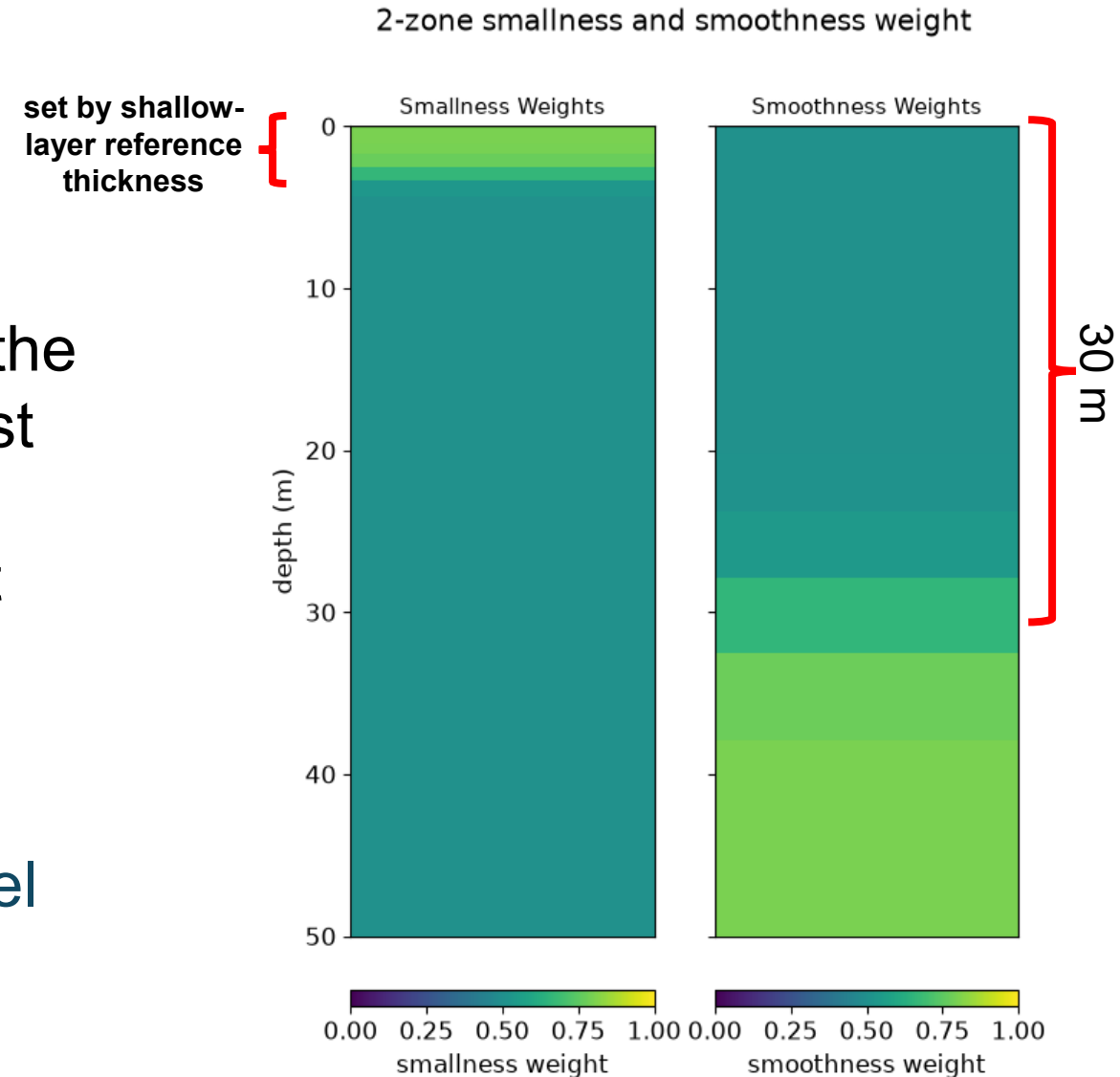
Resistivity (Ohm·m)

Inversion Workflow

2. Regularization: vary smallness and smoothness weights by depth:

- a) **Smallness:** strong near surface (keeps the model close to the reference from the first shallow inversion)
- b) **Smoothness:** enforced more strongly at depth > 30 m (where sensitivity is lower)

3. Final Step: Full-depth L2 inversion with weights and set reference and starting model

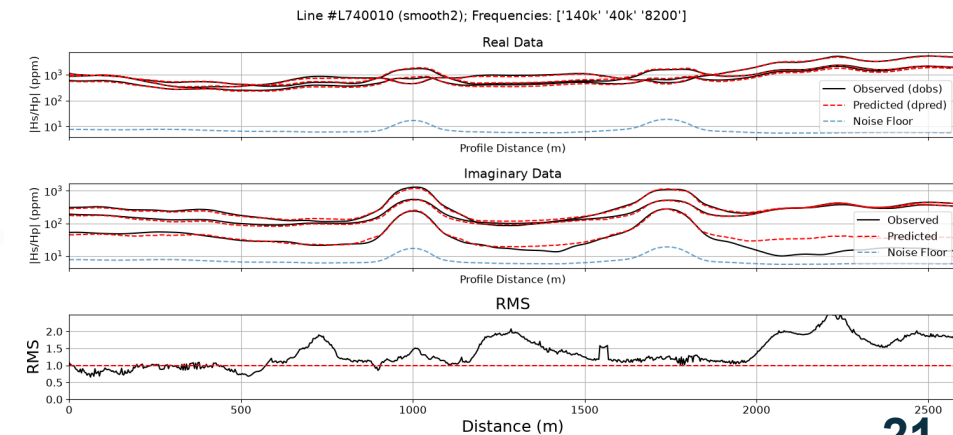
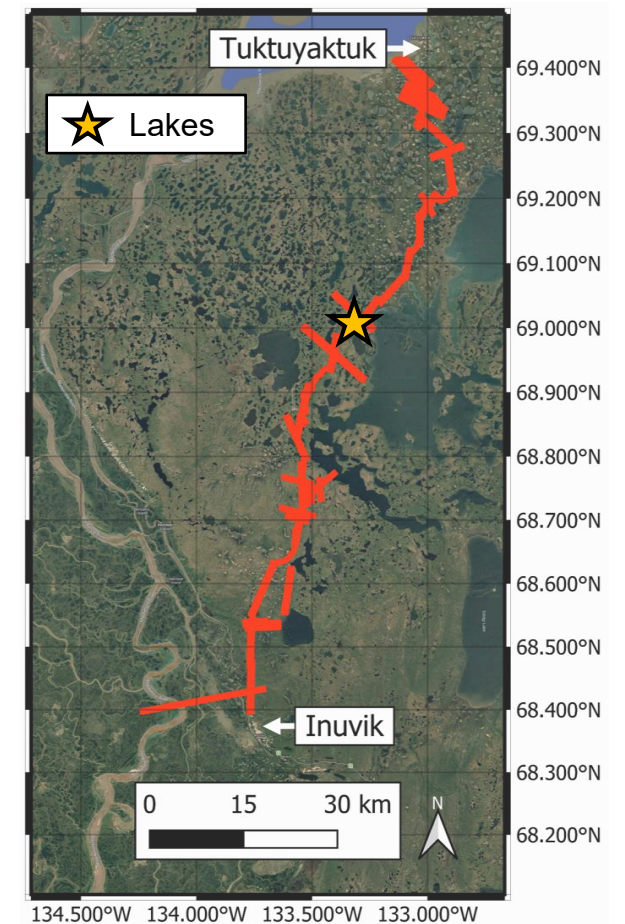
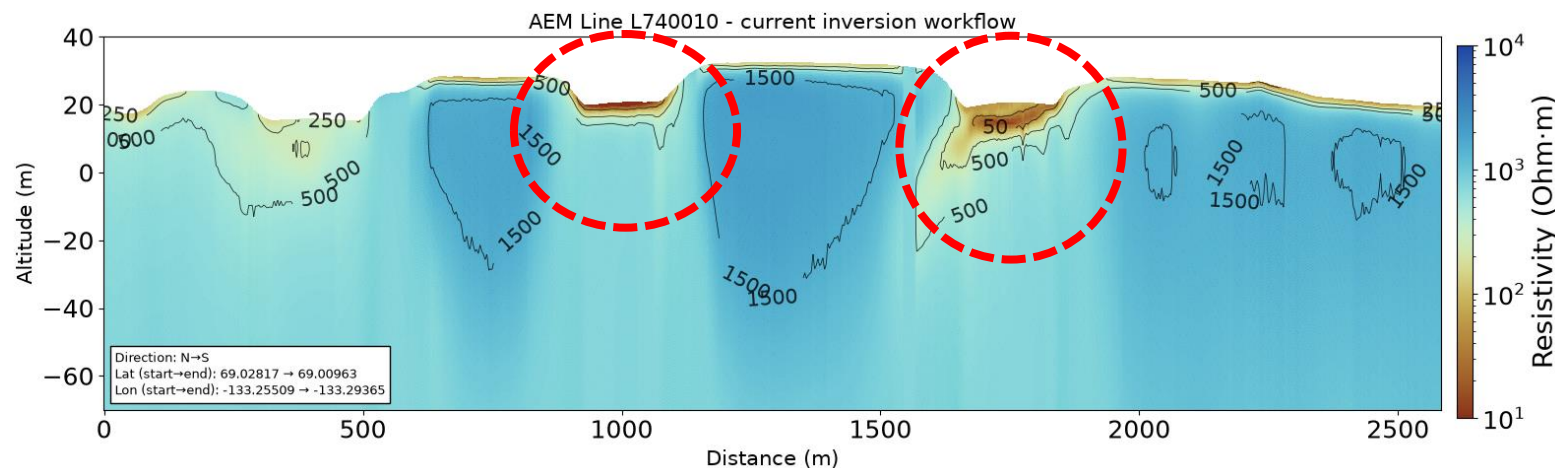


Inversion Workflow

Benefits

- ▶ **Better definition of water bodies and shallow structures; smoother deeper layers with no deep artifacts**
- ▶ **Standardized workflow** — performed consistently across settings, from highly resistive to more conductive, without per-section tuning

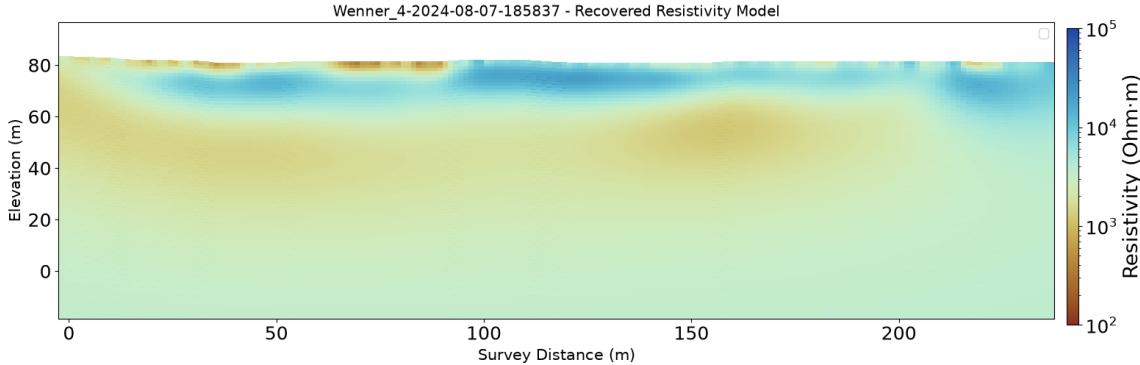
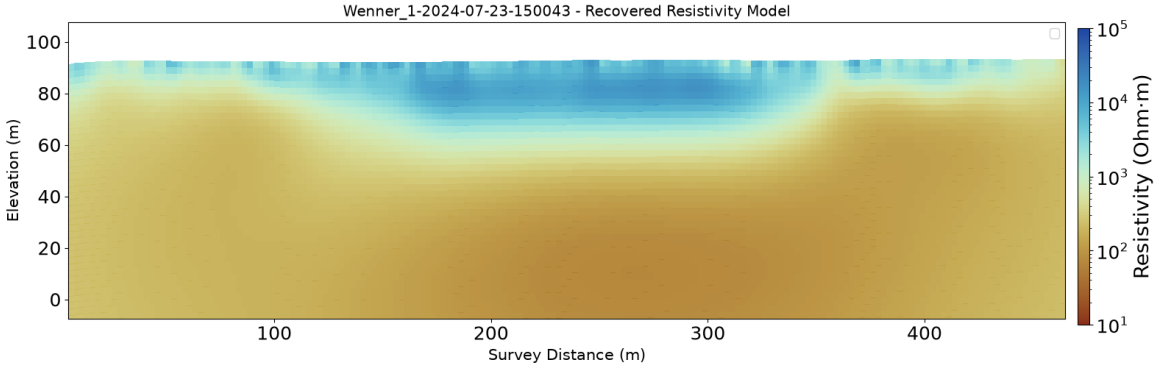
AEM Model — "Lakes" Location



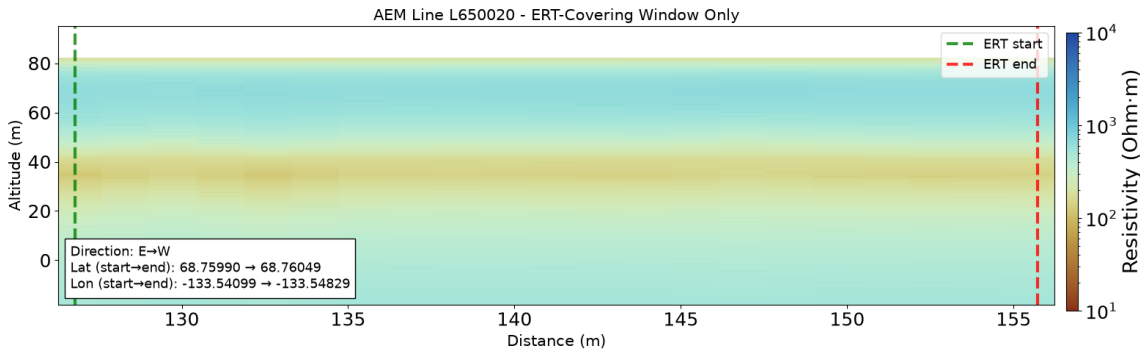
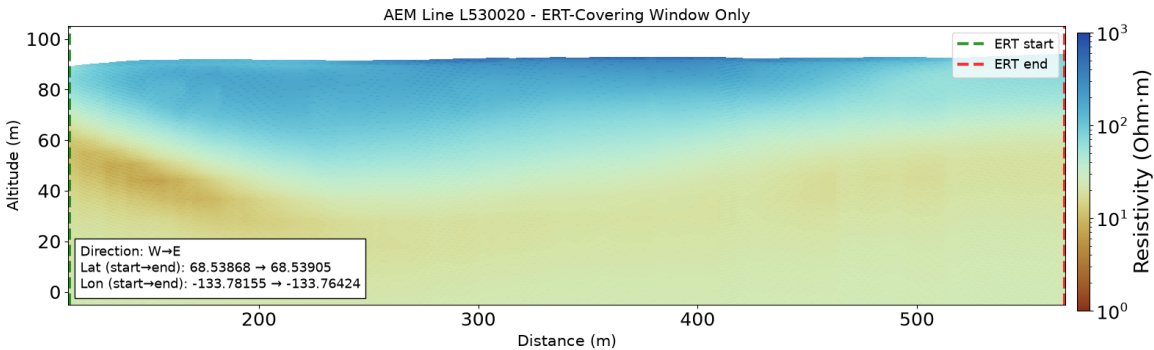
Compare and Validate with ERT models



ERT Lines



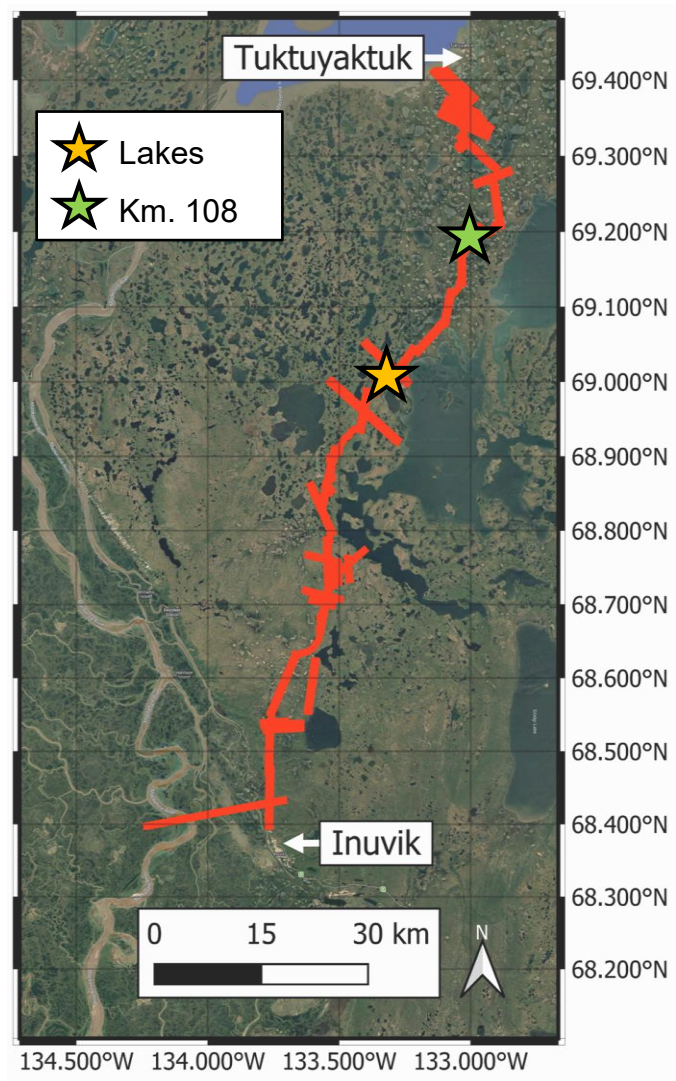
AEM Lines



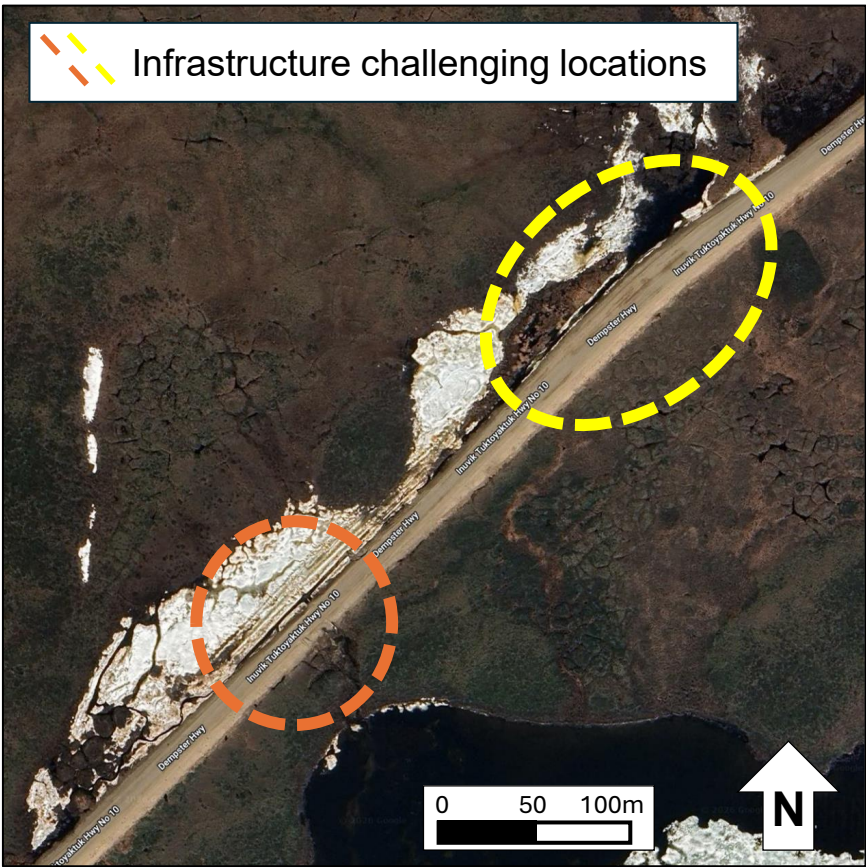
* Different scales — compare overall shape/trends, not absolute values, since ERT and AEM have different sensitivities.

AEM Application for an Infrastructure Problem (Km. 108)

Survey Map with Km. 108



Satellite Imagery to Corroborate Ice/Water Content

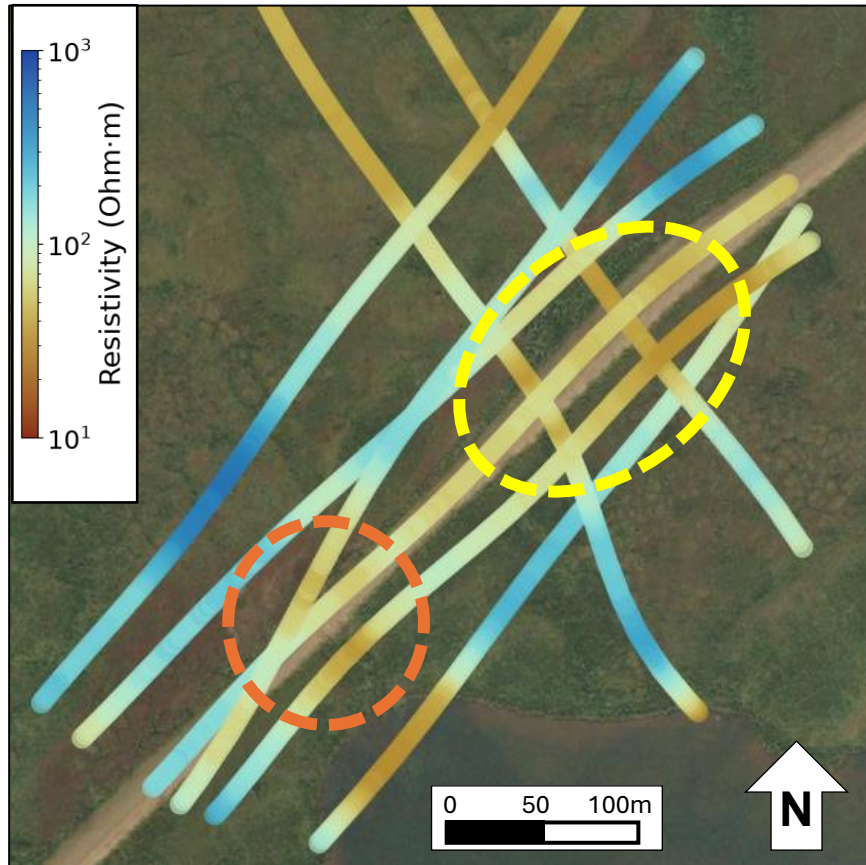


Pictures from locations

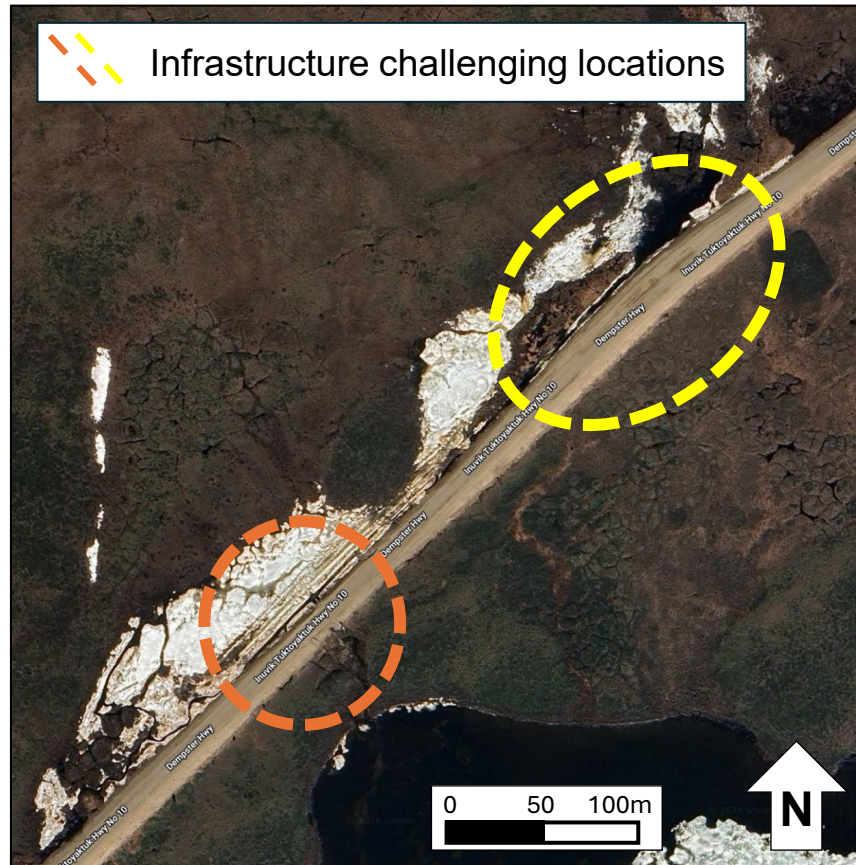


AEM Application for an Infrastructure Problem (Km. 108)

AEM Model Slice (~1 m deep)



Satellite Imagery to Corroborate Ice/Water Content



Pictures from locations



Use shallow AEM model results to identify conductive zones (signature of thaw or water-flow)
Conductive zones → thawed ground → faster permafrost degradation → infrastructure problems

Hans Creek Bridge (HCB) - background

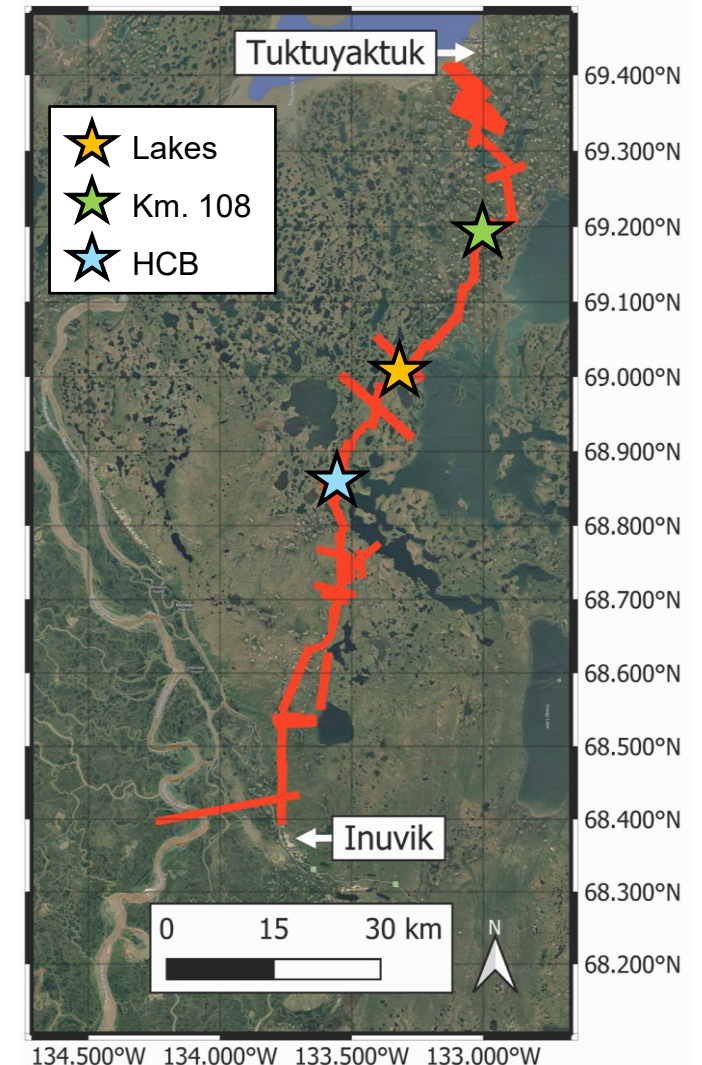
Bridges: concentrated thermal disturbance, high vulnerability.

HCB: adfreeze pile foundation, **requires ground $< -1^{\circ}\text{C}$ for structural stability**

- Expected to be okay for ~35 years. But monitoring shows it could be **reached after only 9 years, requiring intervention**

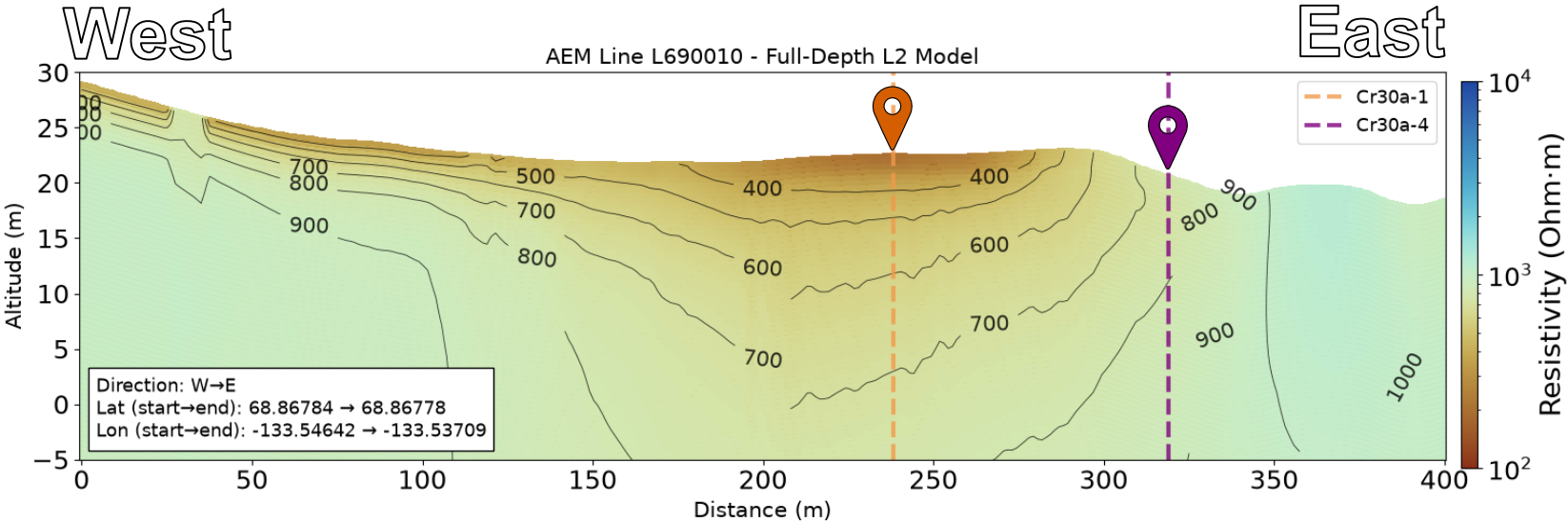
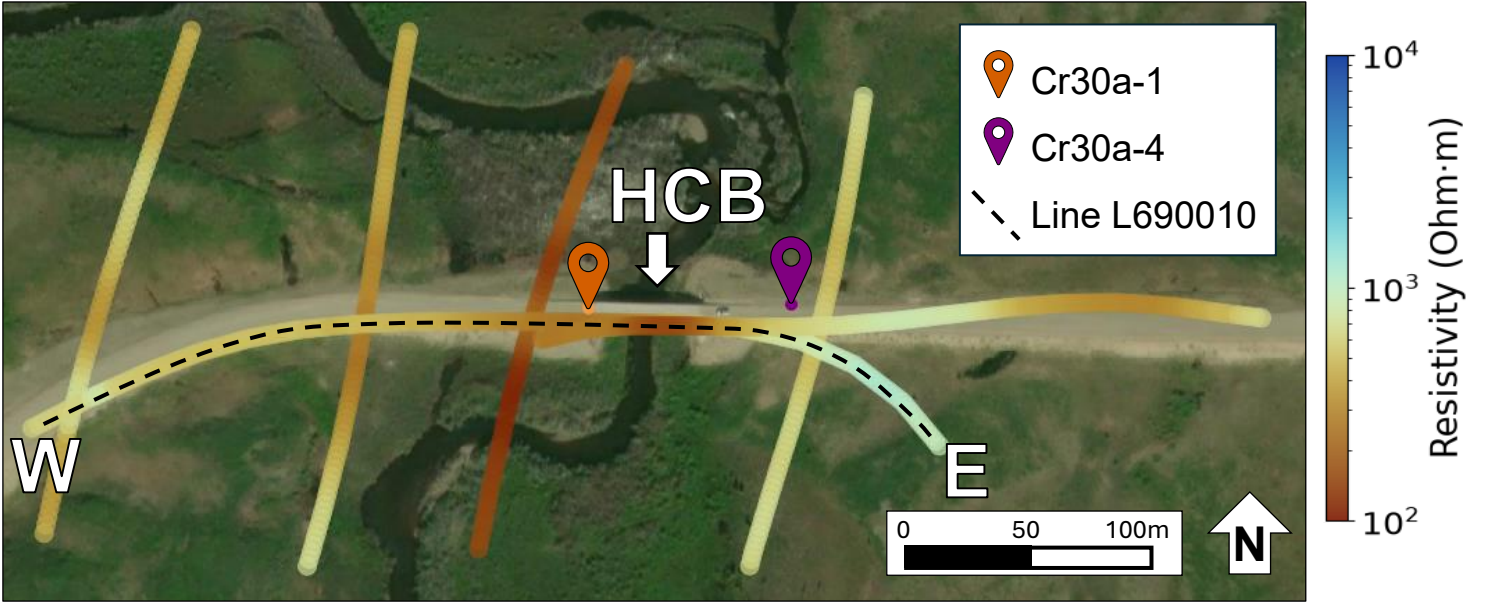
NTGS temperature-monitoring boreholes installed at the site (with temporal and spatial gaps)

Survey Map with HCB

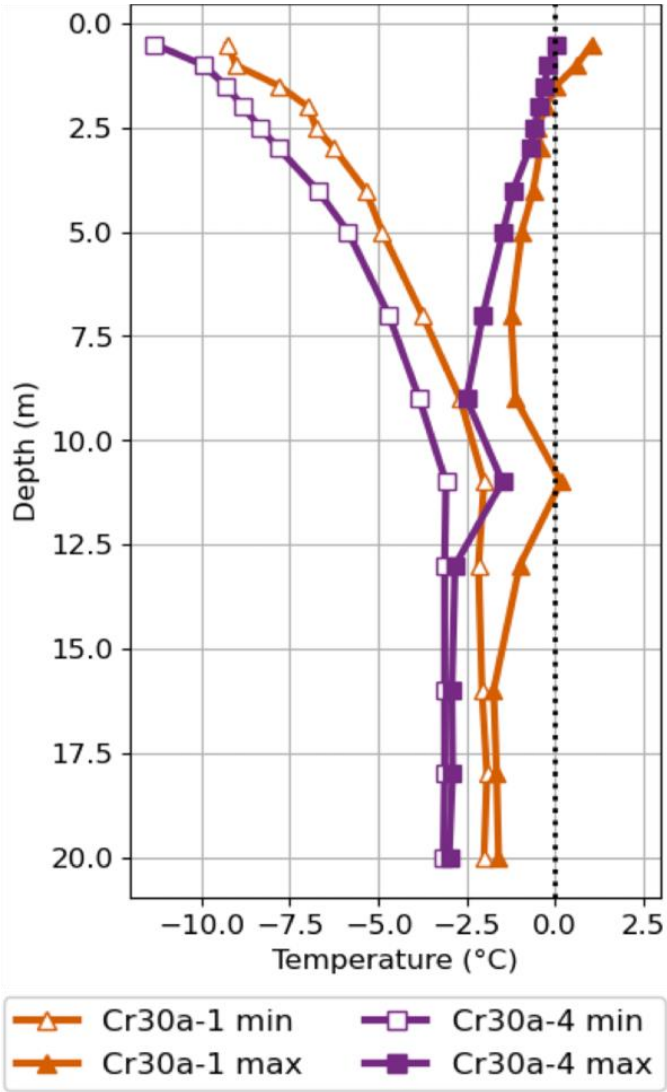


Hans Creek Bridge (HCB)

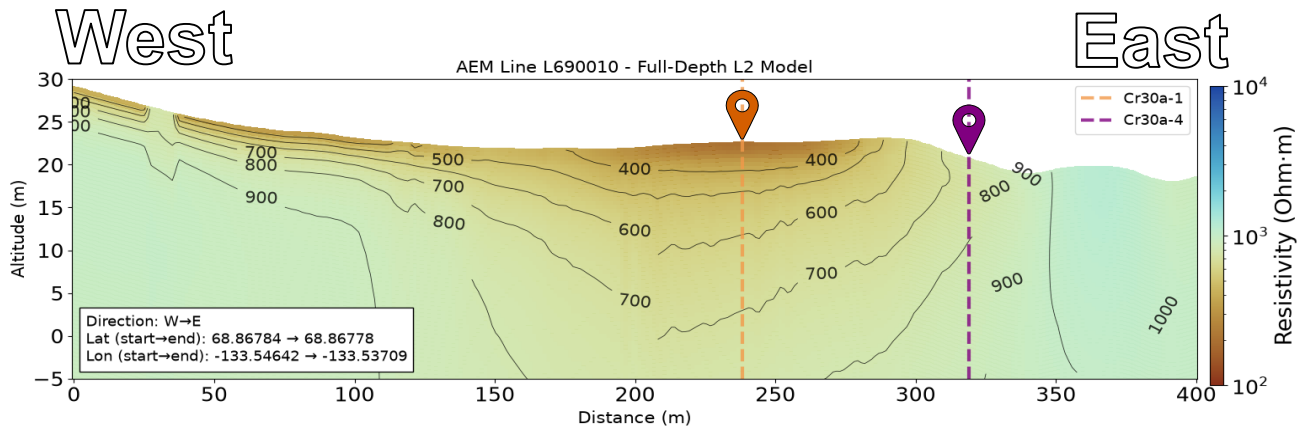
AEM Model Slice (~2m deep)



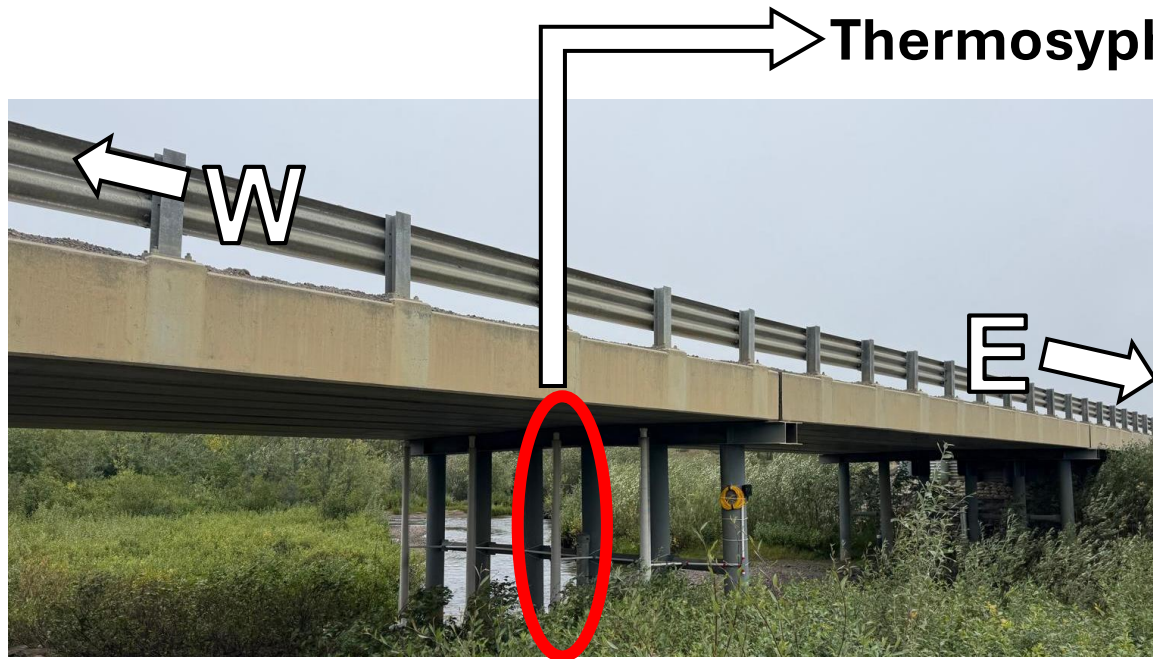
Borehole temperature data up to 2021*



Hans Creek Bridge (HCB)



- **West side more conductive than East**, agrees with limited temperature data (west warmer)
- **Intervention: thermosyphons installed on the West side**



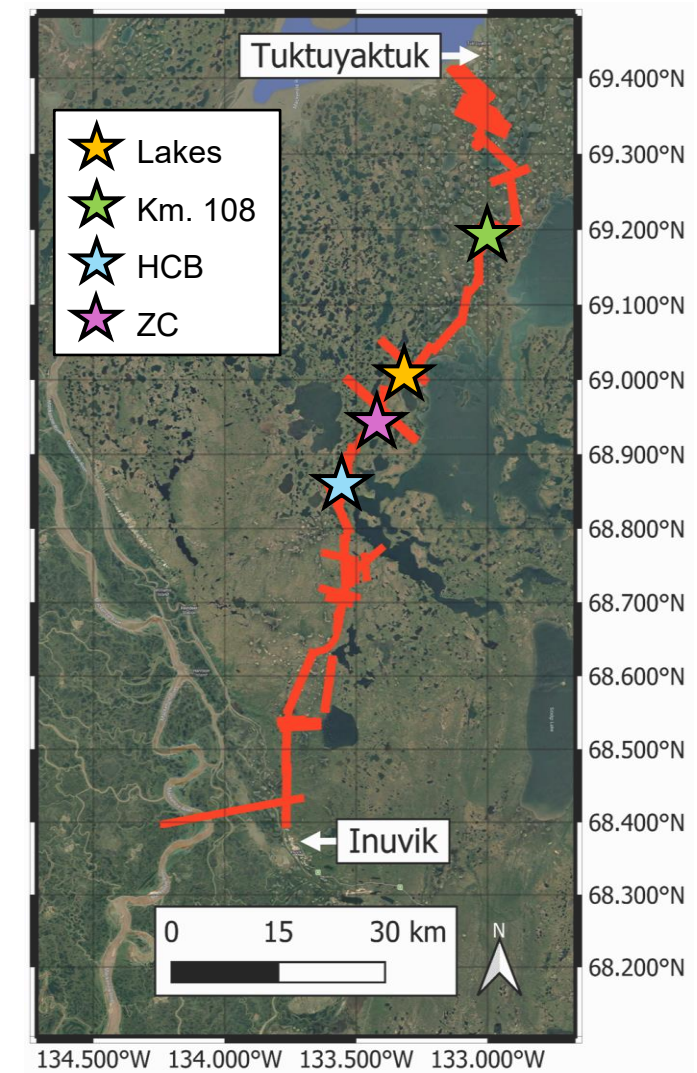
* Passive heat-transfer devices that use natural convection and phase change to cool the ground or structures

Zed Creek Area (ZC) - background

ITH construction altered local ZC stream path.

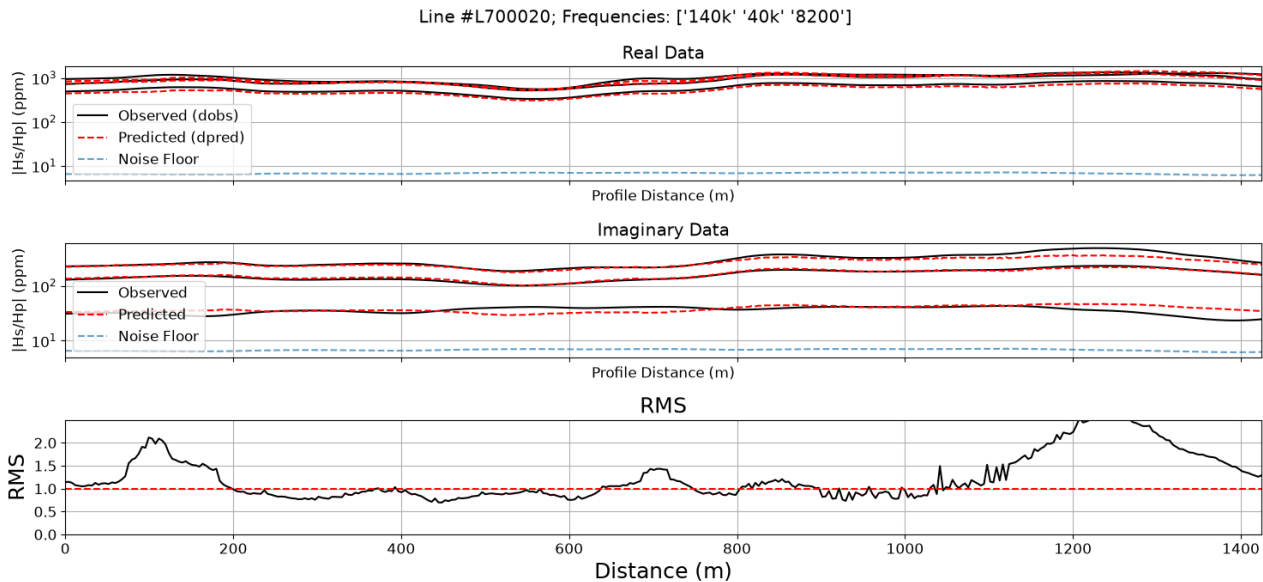
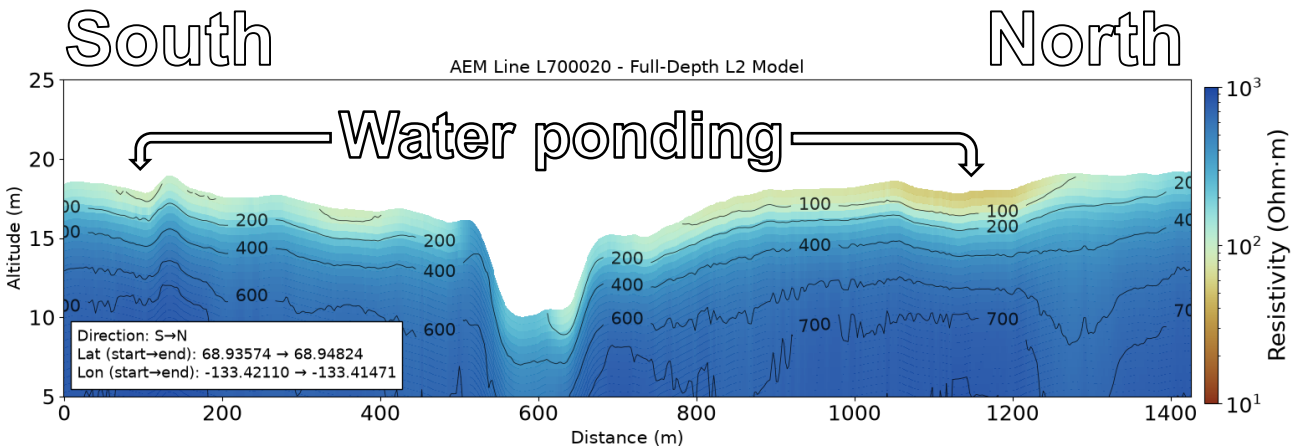
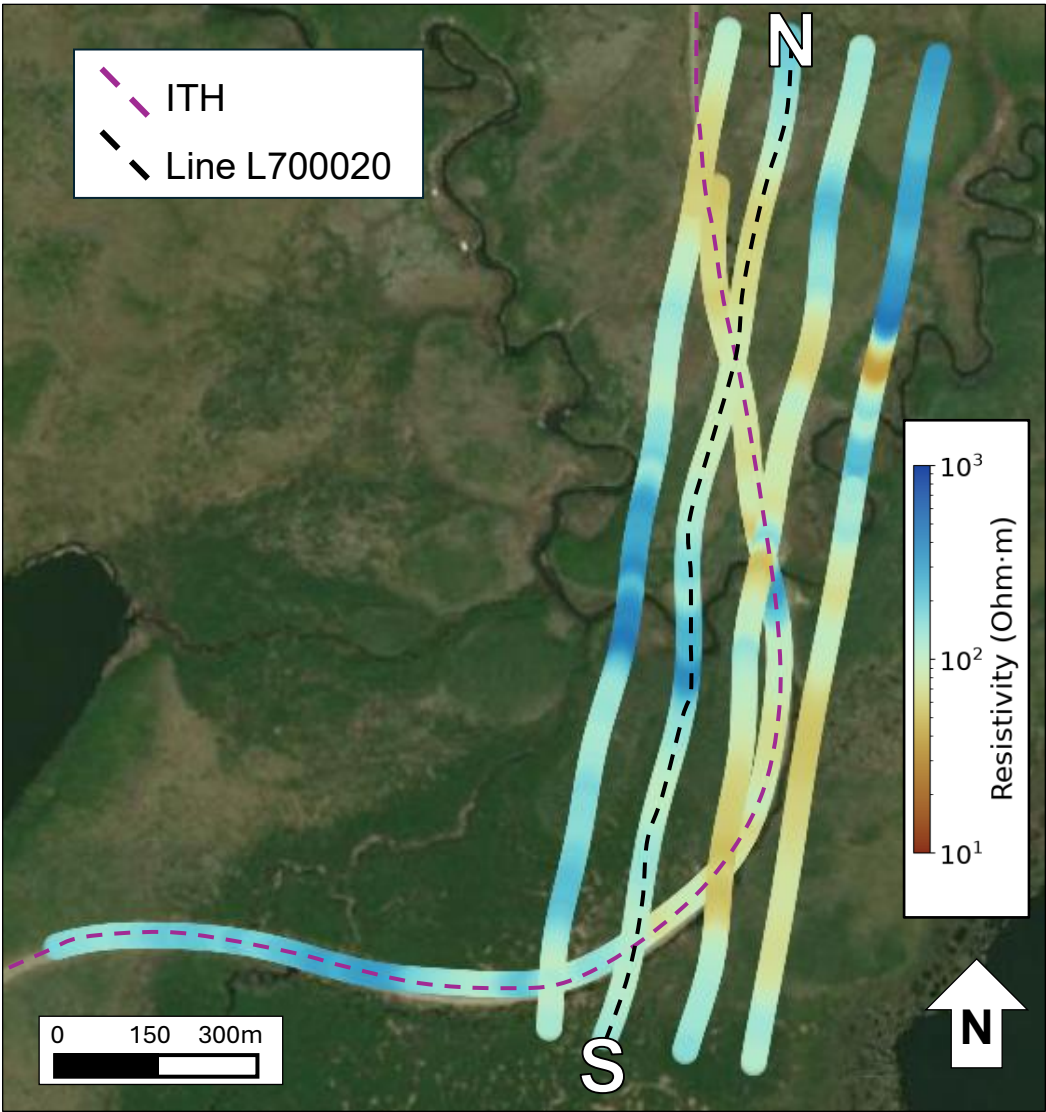
- ▶ Evidence from ERT, LiDAR, and direct probing
- ▶ Visible ponding and flow rerouting, **causing embankment and environmental problems**

Survey Map with ZC



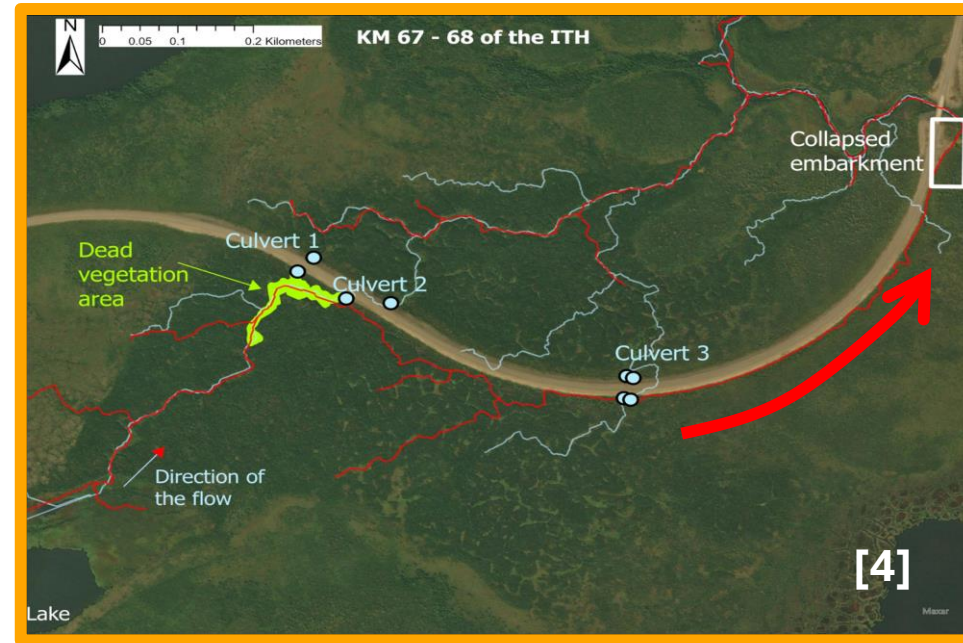
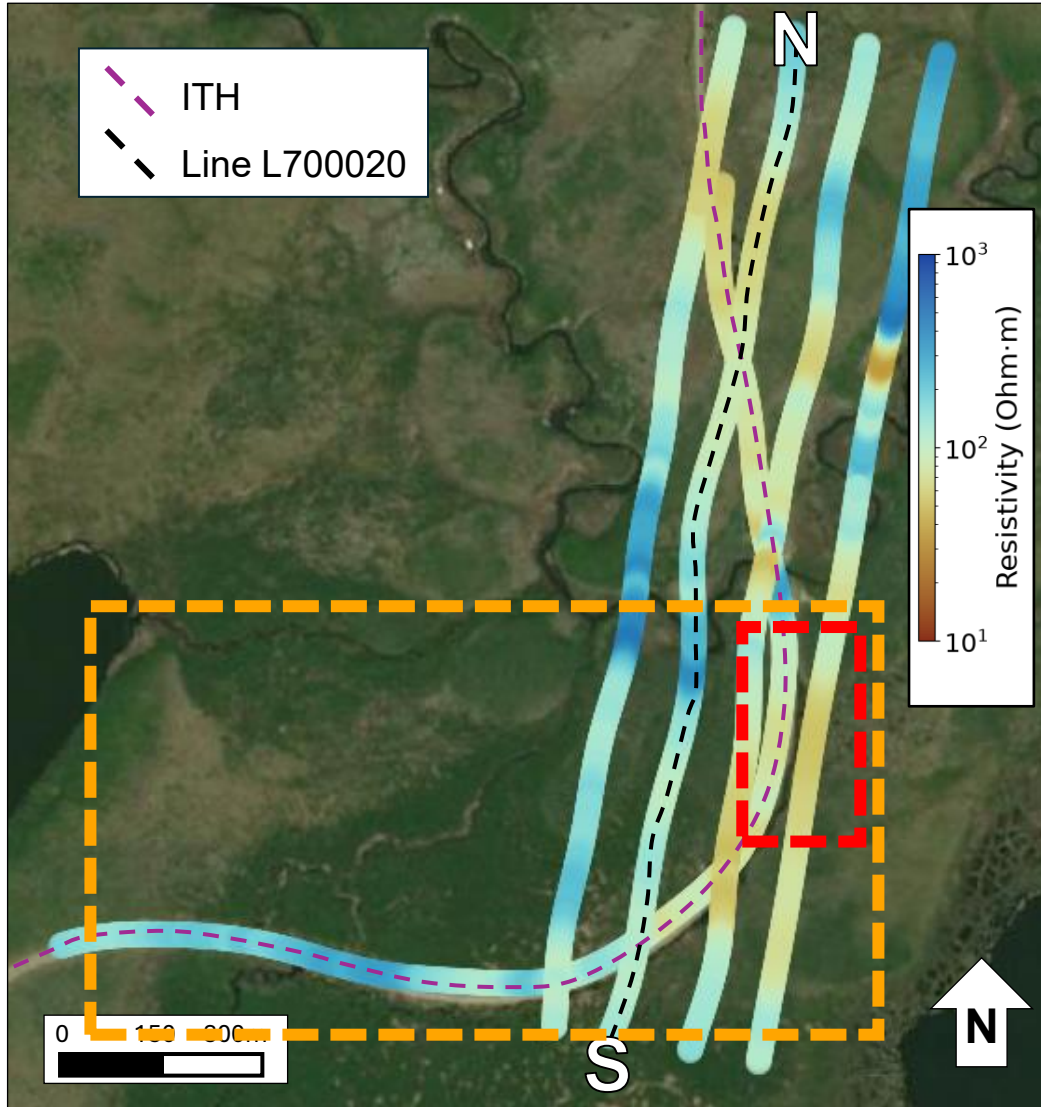
Zed Creek Area (ZC)

AEM Model Slice (~2m deep)



Zed Creek Area (ZC)

AEM Model Slice (~2m deep)



Red lines: new water flow path after road construction.

Ponding area formed along the red arrow.



Overall Results

- ▶ **Northern communities face a growing infrastructure threat** from permafrost degradation, requiring fast, effective characterization methods
- ▶ **AEM provides usable evidence to identify permafrost features that threaten infrastructure**
 - Ex. meaningful evidence supporting the Hans Creek Bridge intervention
- ▶ **Developed an inversion framework tailored to this permafrost setting's data challenges and application.**



Future Work

- ▶ Try other regularization approaches, e.g., laterally constrained inversion
- ▶ Further explore adaptive weighting for inversion regularization
- ▶ Modeling possible induced polarization (IP) effects on the data
- ▶ Explore capability to identify ice-rich zones



Thank you!
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:)



Acknowledgements



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References

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Appendix

Prior EM-Based Permafrost Investigations (Onshore)

Case study	Citation	Survey Type	Survey km	Objective
Northern Margin of the Siberian Platform	Yakovlev et al., 2018	MT + TEM	20,000+ line km (500 m–1 km station spacing)	Image zones of oil and gas accumulation; map permafrost distribution/boundary
Yukon Flats, AK, USA	Minsley et al., 2012	Airborne EM	1,800 line km (300 km ² block + 900 line-km reconnaissance)	Map discontinuous permafrost; demonstrate EM's ability to image permafrost configuration
Illussat, Greenland	Lorentzen et al., 2024	SnowTEM (surface-towed EM)	127 line-km	Map saline permafrost for infrastructure and water-supply concerns

*Content adapted from Gustafson's 2026 EMIW presentation [6]

Other methods...

Beyond our AEM-derived resistivity models, the project integrates:

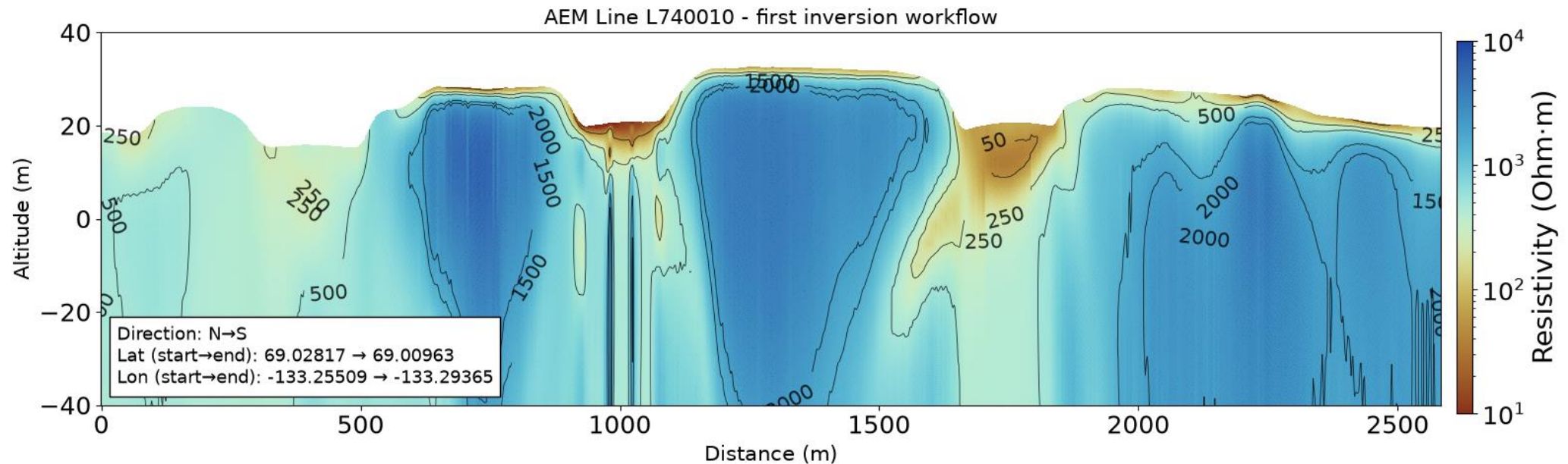
- ▶ Surficial geology mapping and field observations
- ▶ Remote sensing (InSAR, LiDAR, satellite imagery)
- ▶ Electrical Resistivity Tomography (ERT) profiles
- ▶ Borehole data, including ground temperature and moisture measurements

These datasets constrain, validate, and corroborate the AEM-derived models.

First Data Processing and Inversions

Workflow — half-space reference model

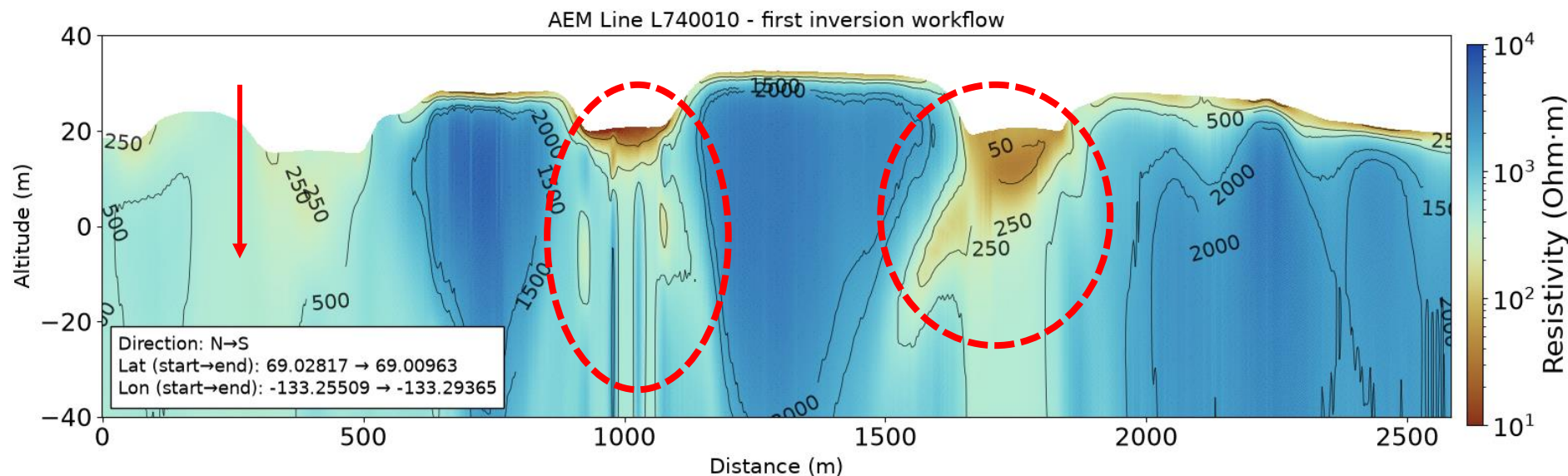
- a) Run half-space inversions for all soundings along each line
- b) Use the mean half-space model as both the starting and reference model for the subsequent 1D, L2-regularized least-squares inversions



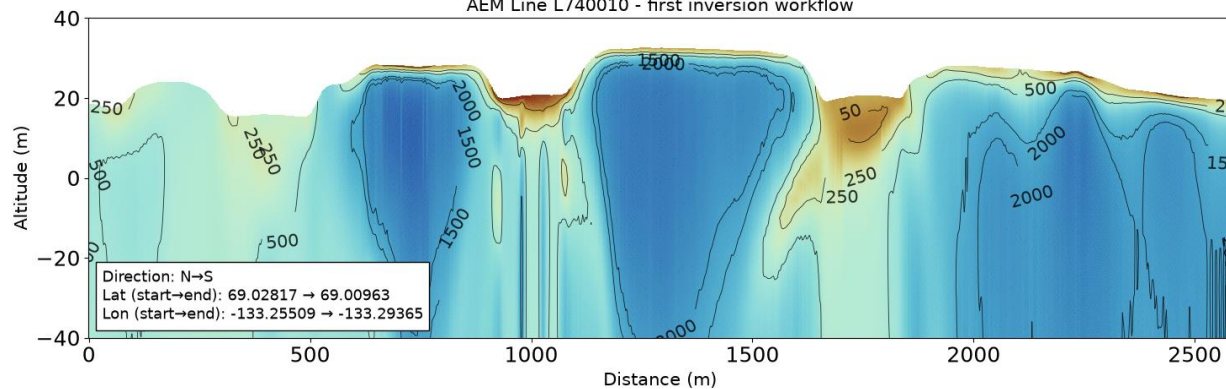
First Data Processing and Inversions

Limitations of the half-space reference approach:

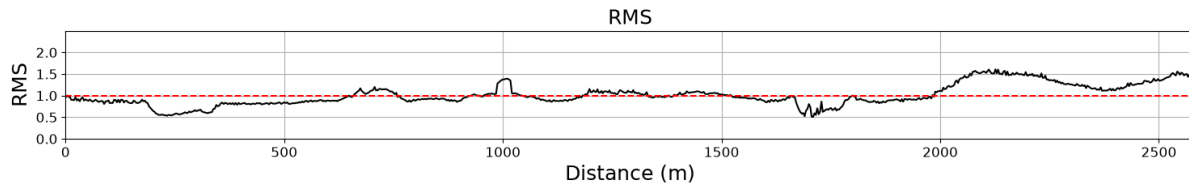
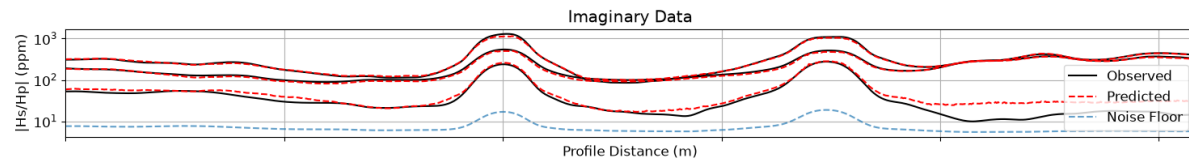
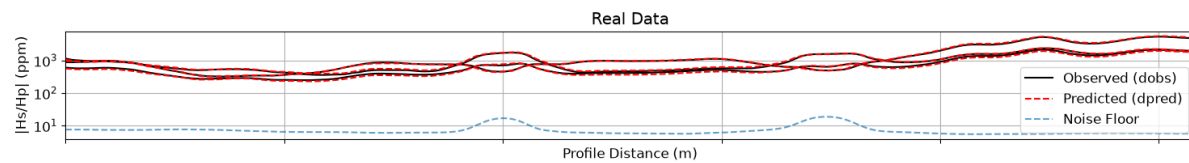
- ▶ **Biases toward deeper structure, stretching shallow structures to greater depths**
- ▶ **Performs worst where feature contrasts are strong laterally**
- ▶ Requires manual tuning for every section — a single half-space reference model doesn't work well along long lines with strongly varying resistivity (e.g., across latitudes)



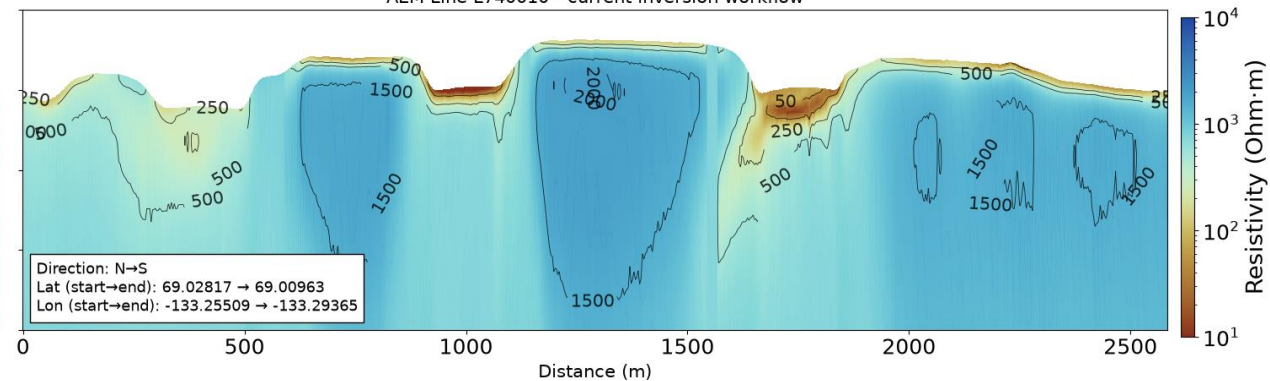
AEM Line L740010 - first inversion workflow



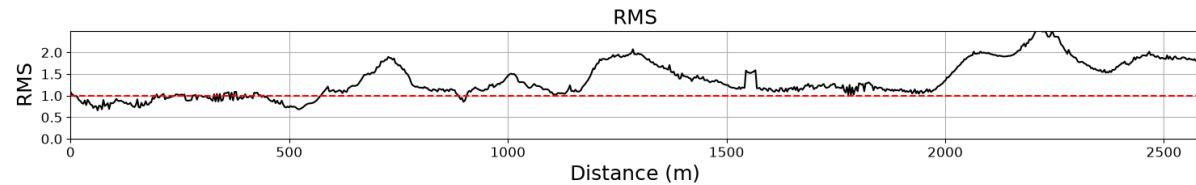
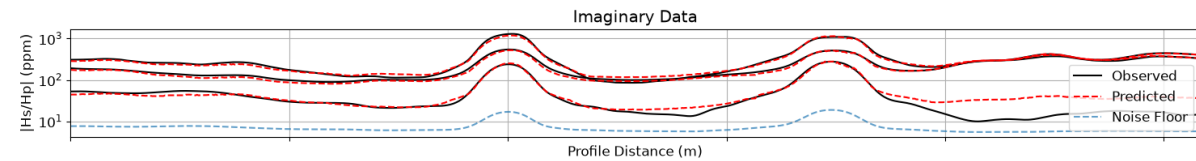
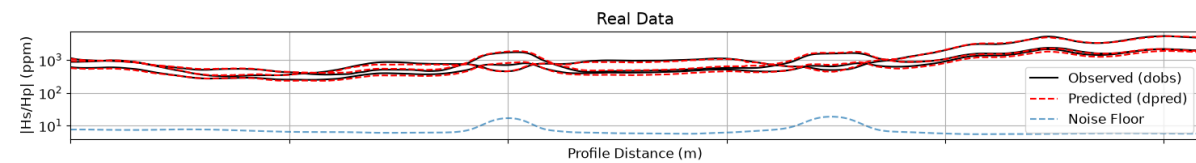
Line #L740010 (normal); Frequencies: ['140k' '40k' '8200']



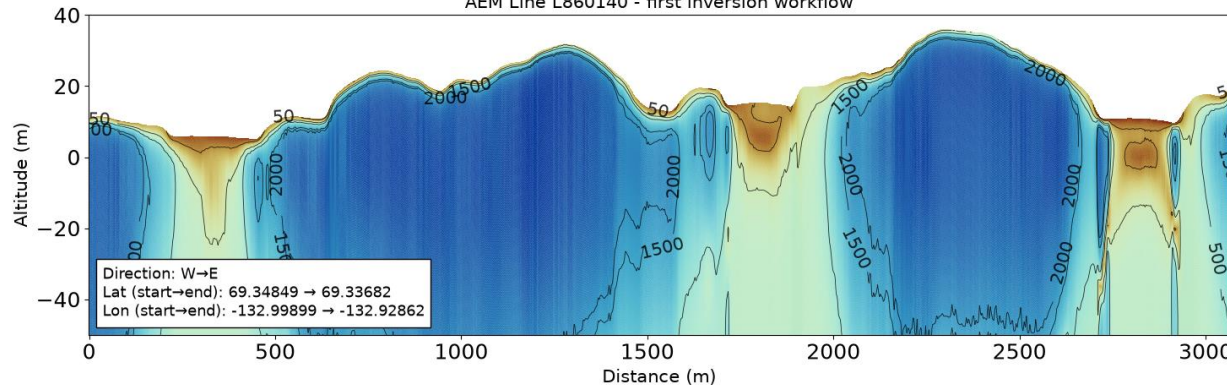
AEM Line L740010 - current inversion workflow



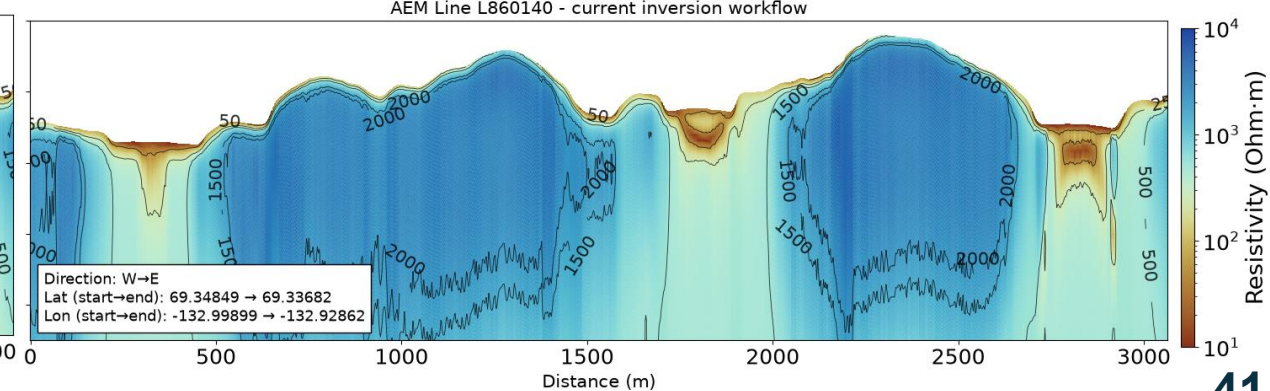
Line #L740010 (smooth2); Frequencies: ['140k' '40k' '8200']



AEM Line L860140 - first inversion workflow

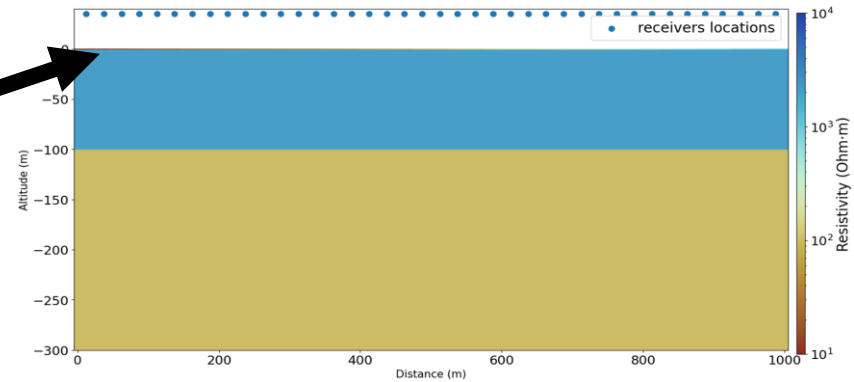


AEM Line L860140 - current inversion workflow

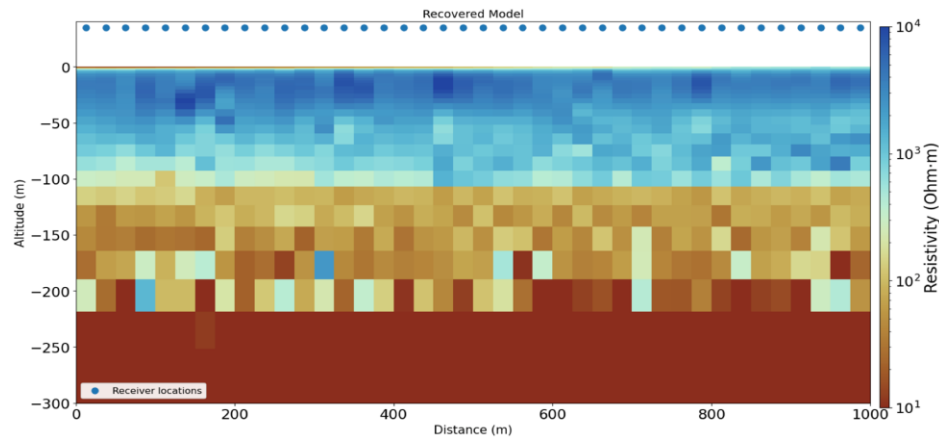


AEM Simulations, Effect of Changing Active Layer

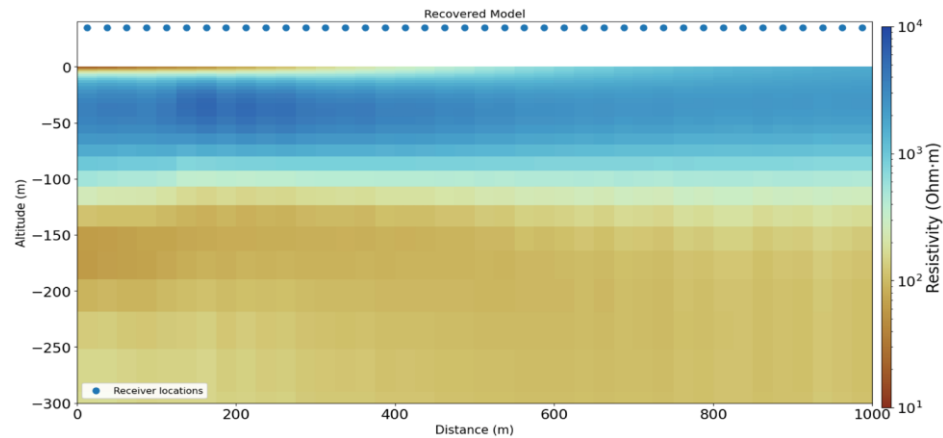
Active Layer (1 m):
10 to 1000 Ohm·m



***With Mean Halfspace Reference!**

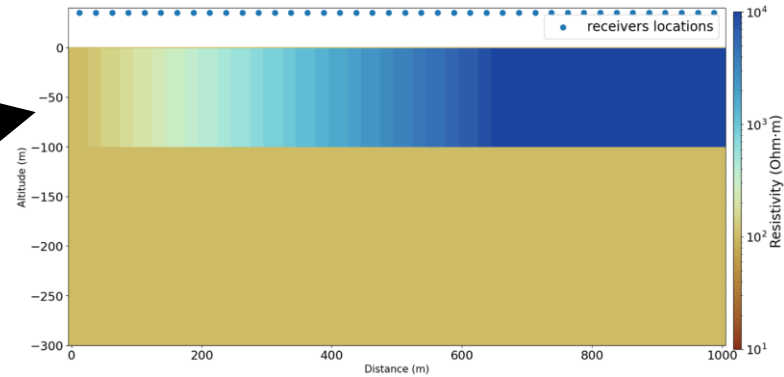


With Mean Active Layer Reference!

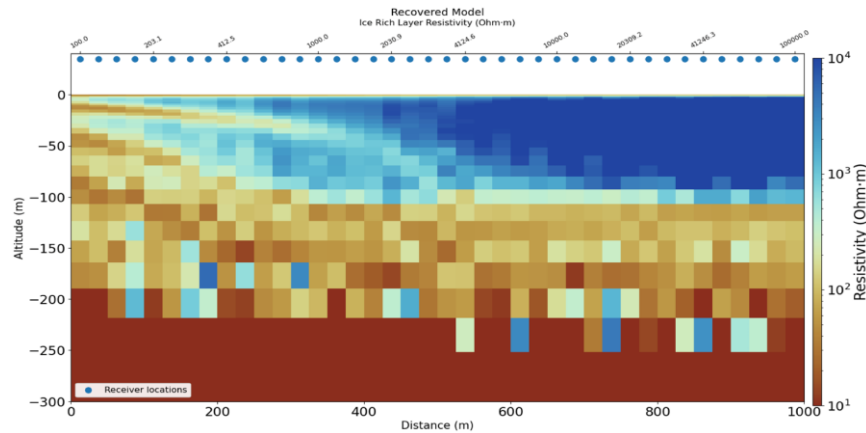


AEM Simulations: Changing Ice Rich Layer

Ice Rich Layer (100 m):
100 to 100'000 Ohm*m



***With Mean Halfspace Reference!**



***With Mean Active Layer Reference!**

